Genome assembly & functional genomics

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Why DNA sequencing?



structural/comparative genomics
genome sequencing

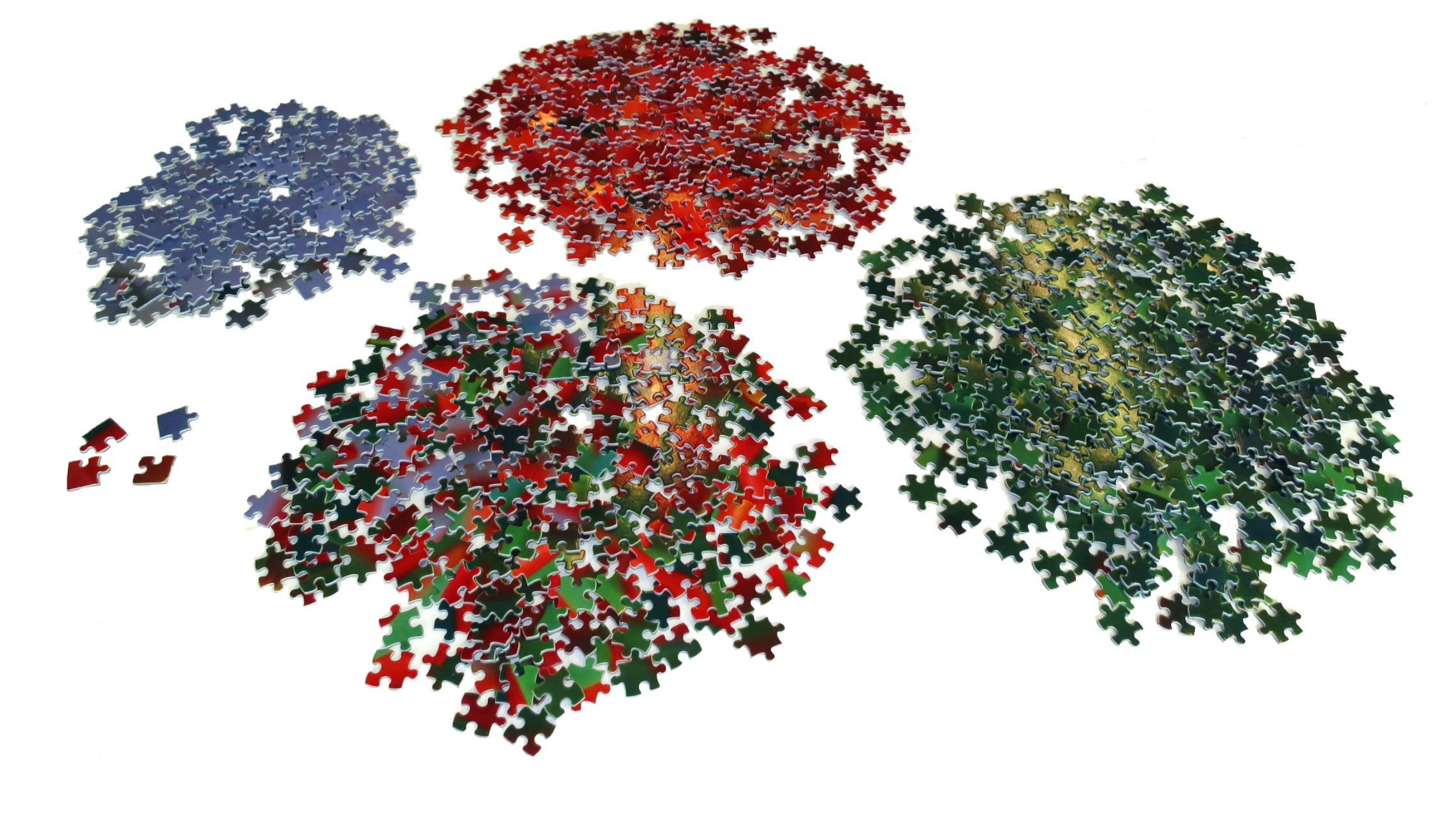
functional genomics:
gene expression analyses, chromatin structure, DNA binding elements

genome diagnostics
human variation (SNPs, structural)

species diagnostics
metagenomics, eDNA, ancient DNA

synthetic biology
information storage


De novo genome assembly



Genome assembly is still a hard problem

Article

Telomere-to-telomere assembly of a complete human X chromosome

https://doi.org/10.1038/s41586-020-2547-7

Received: 30 July 2019

Accepted: 29 May 2020

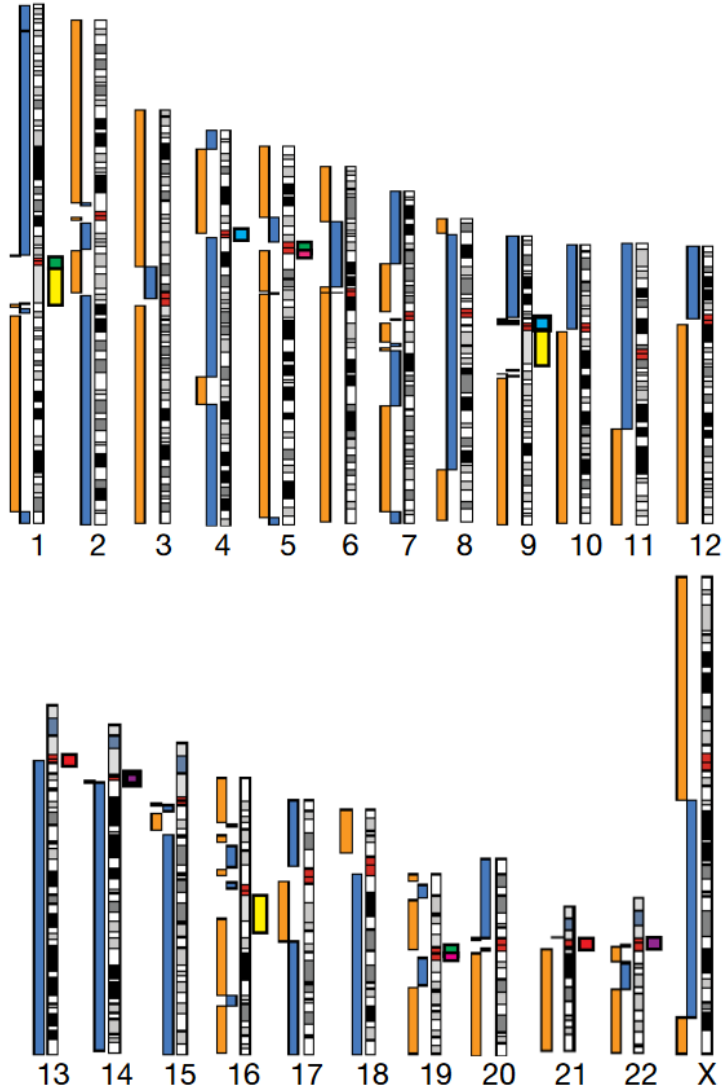
Published online: 14 July 2020

Open access

Check for updates

Karen H. Miga[1,24]✉, Sergey Koren[2,24], Arang Rhie[2], Mitchell R. Vollger[3], Ariel Gershman[4], Andrey Bzikadze[5], Shelise Brooks[6], Edmund Howe[7], David Porubsky[3], Glennis A. Logsdon[3], Valerie A. Schneider[8], Tamara Potapova[7], Jonathan Wood[9], William Chow[9], Joel Armstrong[1], Jeanne Fredrickson[10], Evgenia Pak[11], Kristof Tigyi[1], Milinn Kremitzki[12], Christopher Markovic[12], Valerie Maduro[13], Amalia Dutra[11], Gerard G. Bouffard[6], Alexander M. Chang[2], Nancy F. Hansen[14], Amy B. Wilfert[3], Françoise Thibaud-Nissen[8], Anthony D. Schmitt[15], Jon-Matthew Belton[15], Siddarth Selvaraj[15], Megan Y. Dennis[16], Daniela C. Soto[16], Ruta Sahasrabudhe[17], Gulhan Kaya[16], Josh Quick[18], Nicholas J. Loman[18], Nadine Holmes[19], Matthew Loose[19], Urvashi Surti[20], Rosa ana Risques[10], Tina A. Graves Lindsay[12], Robert Fulton[12], Ira Hall[12], Benedict Paten[1], Kerstin Howe[9], Winston Timp[4], Alice Young[6], James C. Mullikin[6], Pavel A. Pevzner[21], Jennifer L. Gerton[7], Beth A. Sullivan[22], Evan E. Eichler[3,23] & Adam M. Phillippy[2]✉

After two decades of improvements, the current human reference genome (GRCh38) is the most accurate and complete vertebrate genome ever produced. However, no single chromosome has been finished end to end, and hundreds of unresolved gaps persist[1,2].



Earth's heart of iron begins to yield its secrets p. 18

Microglia in chronic pain recovery and relapse pp. 33 & 86

Particle acceleration in a nova explosion p. 77

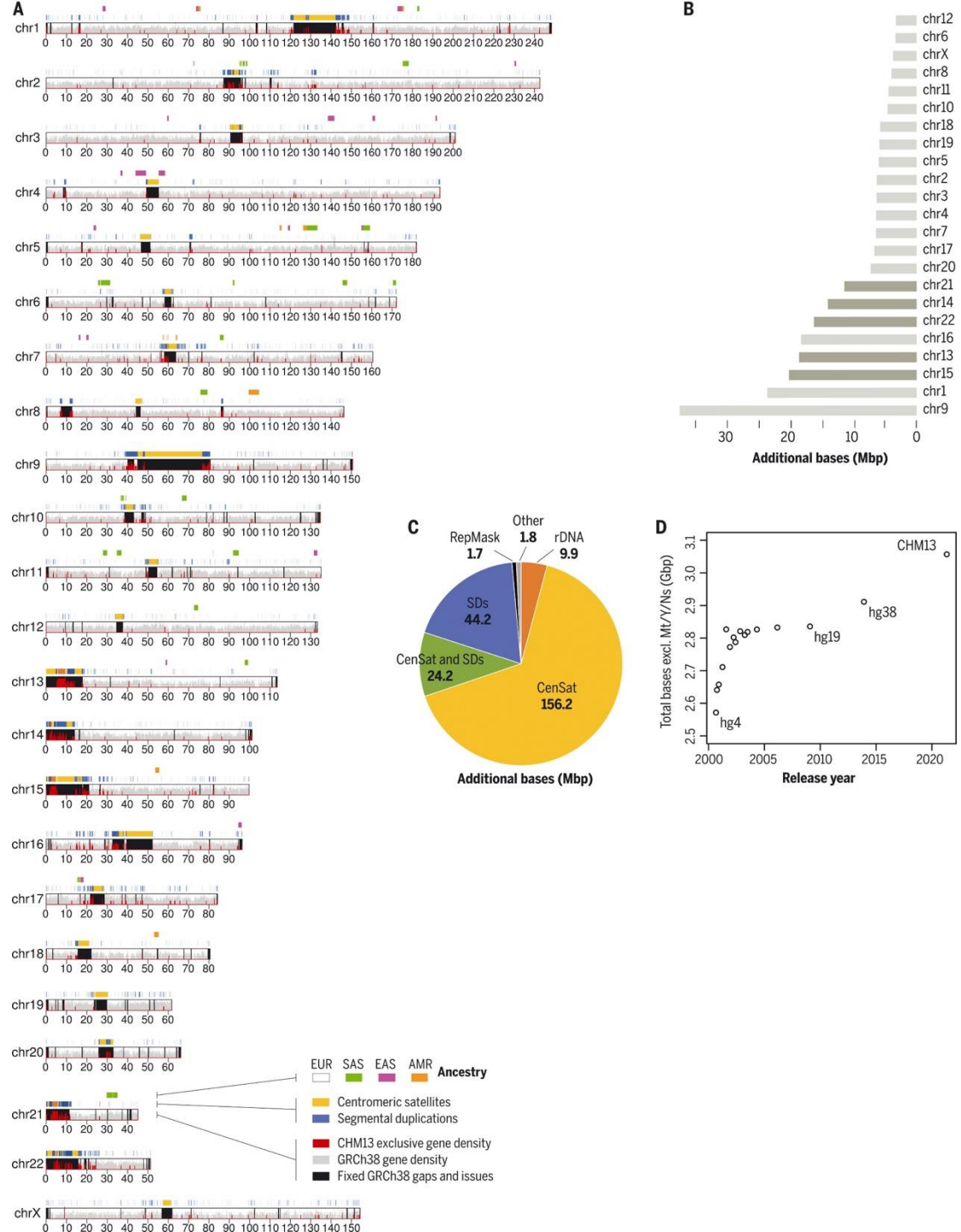
Science

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FILLING THE GAPS

Closing in on a complete human genome p. 42





The first problem: sequencing limitations

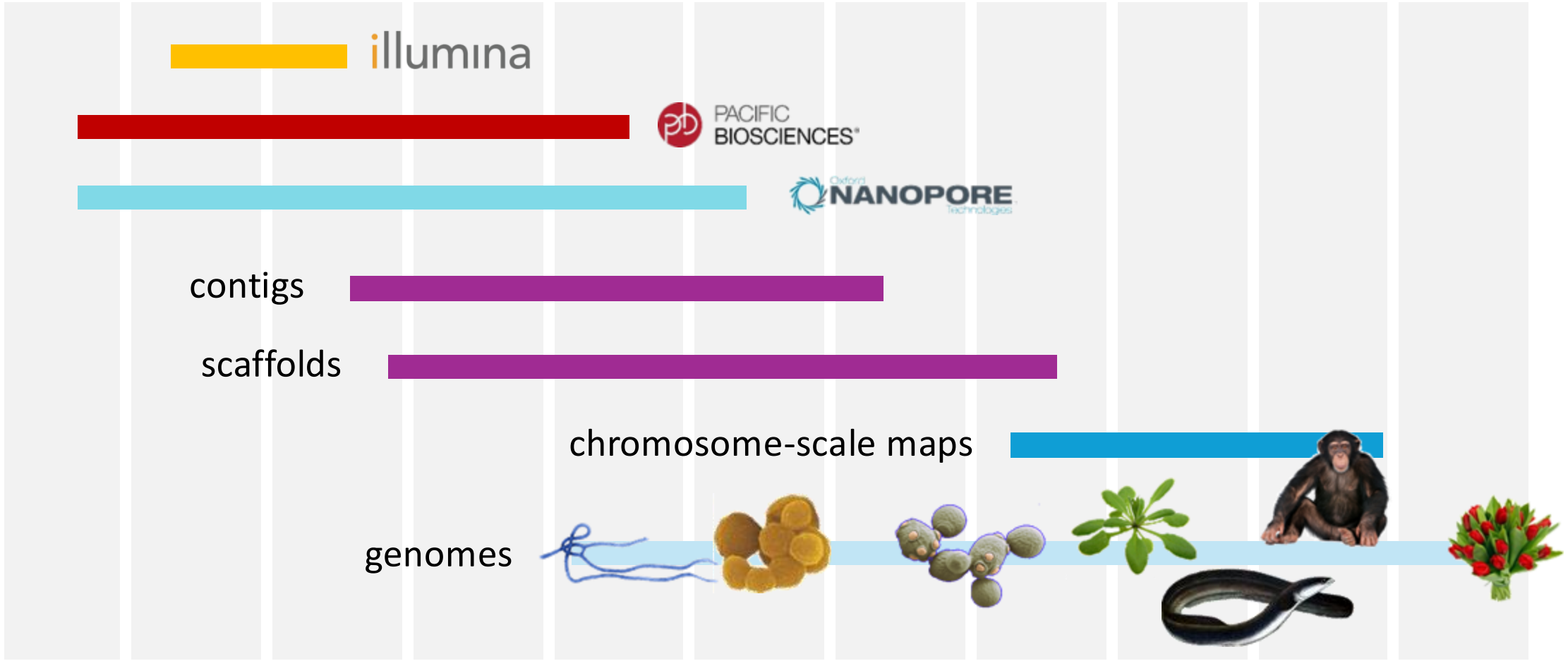


The second problem: technical errors



The third problem: genomic architecture

Genome sequencing



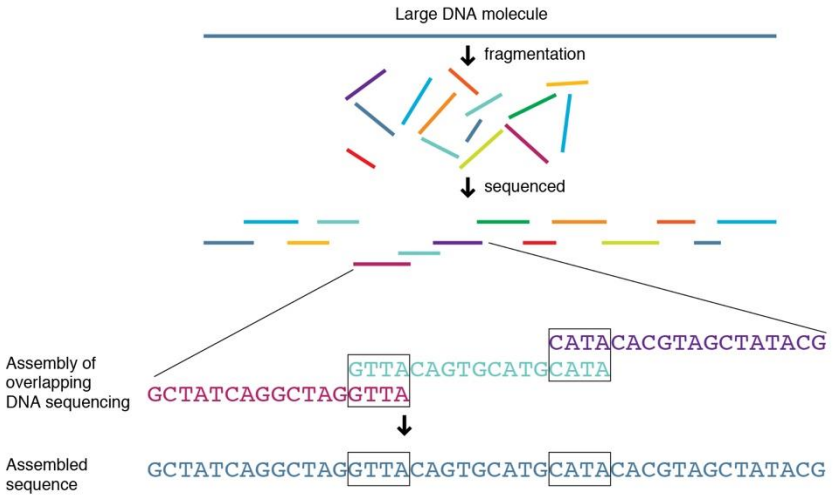
The first genome assembly problem

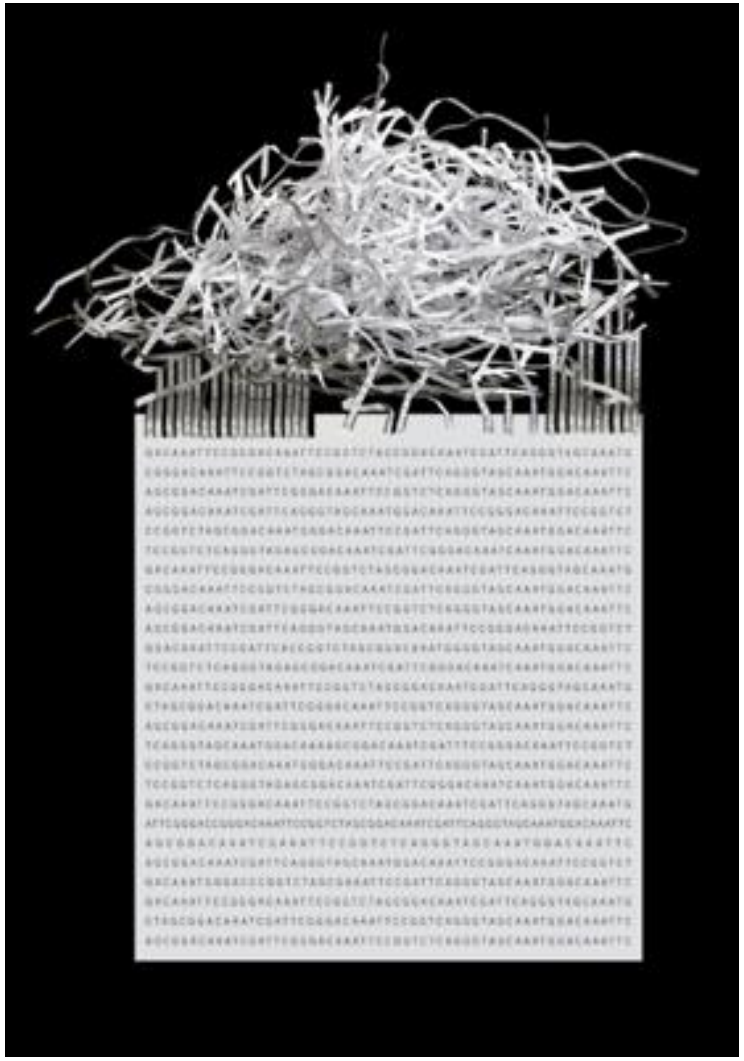
- Genomes are very long
- Sequencing reads are relatively short
- Isolating full chromosome DNA molecules is almost impossible (because of strand breaks)
- How can we reconstruct a very long sequence from very short reads?



Shotgun genome assembly

- Bottom-up approach
  - generate random small fragments from the entire genome
  - sequence many fragments (every position >8-50×)
  - let the computer sort it out based on overlapping reads
  - entire process is relatively fast and cheap
  - assembly is error-prone and computationally challening





Genome sequencing



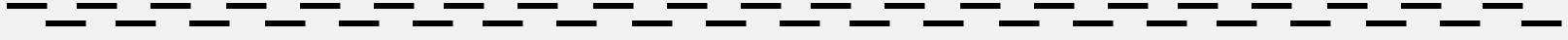
in vivo

genome

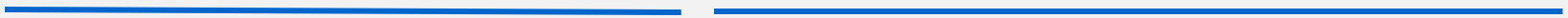


in vitro

library



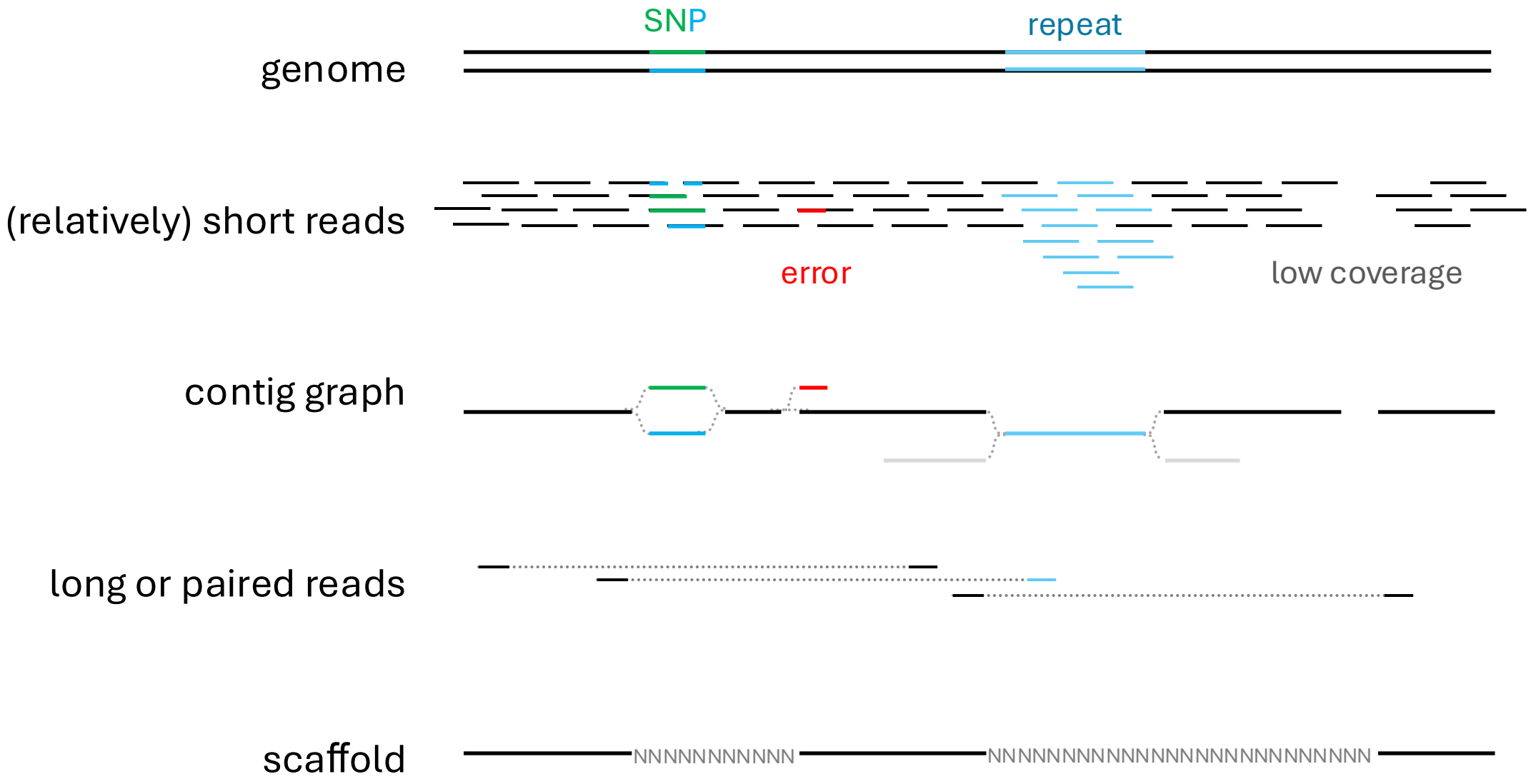
reads



in silico

de novo assembly

From genome to contigs and scaffolds



Assembly of reads

Different tools/algorithms:

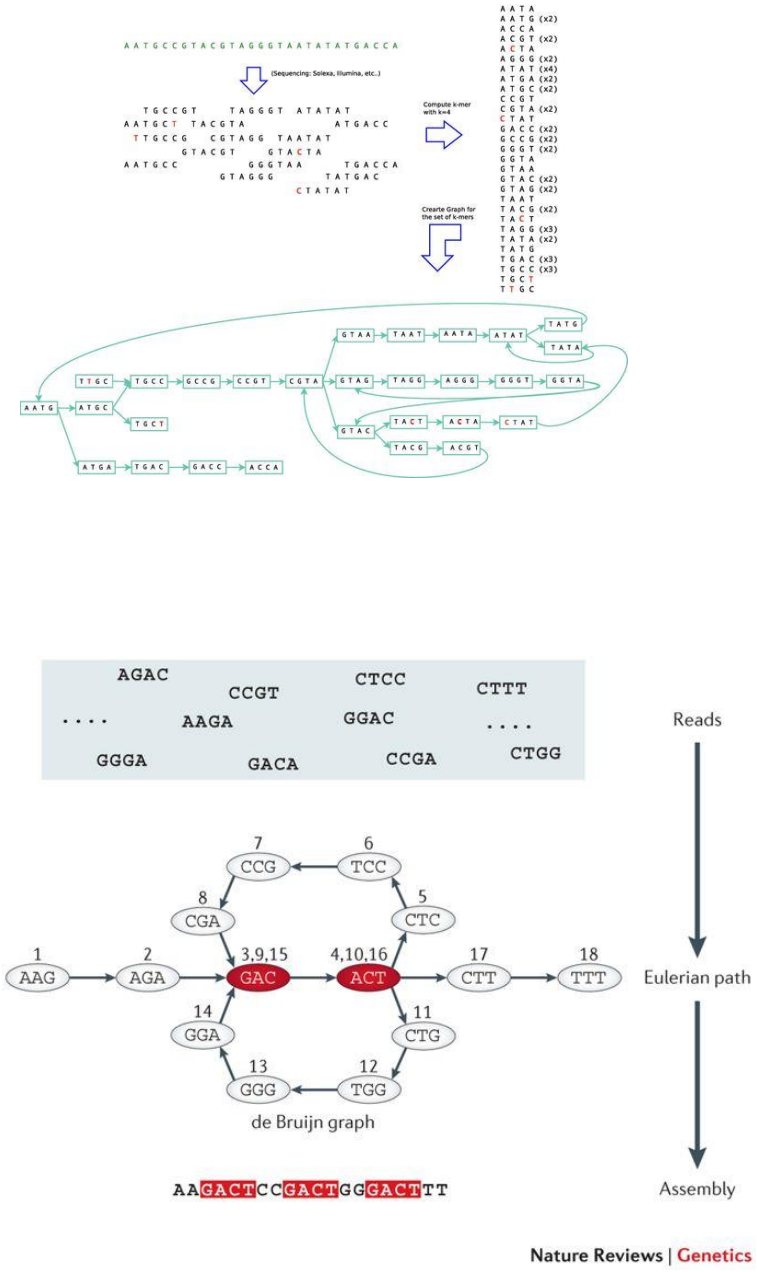
Assembler (acronym)
AbySS (ABYSS)
Celera (CELERA)
CLC Assembly Cell (CLC)
GS De Novo Assembler (NEWBLER)
MIRA (MIRA)
SeqMan Ngen (SEQMAN)
SOAPdenovo2 (SOAP2)
SPAdes (SPADES)
Velvet (VELVET)

PLOS: Jünemann et al., 2014. GABenchToB: A Genome Assembly Benchmark Tuned on Bacteria and Benchtop Sequencers. PLoS ONE 9(9): e107014.

https://github.com/nadegeguiglielmoni/genome_assembly_tools/

assemblers

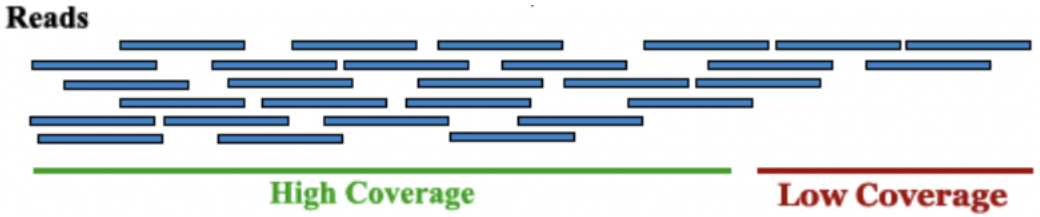
technology	name	link	publication	last_update
Sanger	ARACHNE		10.1101/gr.208902	
Sanger	Atlas	https://www.hgsc.bcm.edu/software/atlas-2	10.1101/gr.2264004	2013
Sanger	CAP3	https://faculty.sites.iastate.edu/xqhuang/cap3-assembly-program	10.1101/gr.9.9.868	
Sanger	Celera		10.1093/bioinformatics/btn074	
Sanger	Euler		10.1073/pnas.171285098	
Sanger	JAZZ		10.1126/science.1072104	
Sanger	Minimus		10.1186/1471-2105-8-64	
Sanger	phrap		10.1101/gr.8.3.186	
Sanger	Phusion		10.1101/gr.731003	
Sanger	TIGR		10.1089/gst.1995.1.9	
High-accuracy short reads	ABySS	https://github.com/bcgsc/abyss	10.1101/gr.214346.116	2022
High-accuracy short reads	ALLPATHS	https://software.broadinstitute.org/allpaths-lg/blog/?page_id=12	10.1101/gr.7337908	2008
High-accuracy short reads	BASE	https://github.com/dhlbh/BASE	10.1186/s12864-016-2829-5	2017
High-accuracy short reads	CABOG	http://wgs-assembler.sourceforge.net	10.1093/bioinformatics/btn548	2008
High-accuracy short reads	Edena	http://www.genomic.ch/edena.php	10.1101/gr.072033.107	2013
High-accuracy short reads	EPGA	https://github.com/bioinformaticsCSU/EPGA	10.1093/bioinformatics/btu762	2017
High-accuracy short reads	Euler-SR	http://web.archive.org/web/20110720080755/http://euler-assembler.ucsd.edu/euler-sr.1.1.2.tgz	10.1101/gr.7088808	2011
High-accuracy short reads	Gossamer		10.1093/bioinformatics/bts297	2012
High-accuracy short reads	IDBA	https://github.com/loneknightpy/idba	10.1007/978-3-642-12683-3_28	2016
High-accuracy short reads	ISEA		10.1109/TCBB.2016.2550433	
High-accuracy short reads	JR-Assembler		10.1073/pnas.1314090110	
High-accuracy short reads	LightAssembler		10.1093/bioinformatics/btw470	
High-accuracy short reads	Meraculous		10.1371/journal.pone.0023501	
High-accuracy short reads	Minia	https://github.com/GATB/minia	10.1186/1748-7188-8-22	2022
High-accuracy short reads	Mira		10.1.1.23.7465	
High-accuracy short reads	Newbler			
High-accuracy short reads	PCAP	https://faculty.sites.iastate.edu/xqhuang/pcap-assembly-program	10.1101/gr.1390403	
High-accuracy short reads	PE-Assembler		10.1093/bioinformatics/btq626	
High-accuracy short reads	PERGA		10.1371/journal.pone.0114253	
High-accuracy short reads	Platanus		10.1101/gr.170720.113	
High-accuracy short reads	QSRA		10.1186/1471-2105-10-69	
High-accuracy short reads	Ray		10.1089/cmb.2009.0238	
High-accuracy short reads	Readjoiner		10.1186/1471-2105-13-82	
High-accuracy short reads	SGA		10.1101/gr.126953.111	
High-accuracy short reads	SHARCGS		10.1101/gr.6435207	
High-accuracy short reads	SOAPdenovo		10.1101/gr.097261.109	
High-accuracy short reads	SOAPdenovo2		10.1186/2047-217X-1-18	
High-accuracy short reads	SPAdes		10.1089/cmb.2012.0021	
High-accuracy short reads	SparseAssembler		10.1186/1471-2105-13-S6-S1	
High-accuracy short reads	SSAKE		10.1093/bioinformatics/bti629	
High-accuracy short reads	SUTTA		10.1093/bioinformatics/btq646	
High-accuracy short reads	VCAKE		10.1093/bioinformatics/btm451	
High-accuracy short reads	Velvet		10.1002/0471250953.bi1105s31	
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Low-accuracy long reads	Canu	https://github.com/marbl/canu	10.1101/gr.215087.116	2021
Low-accuracy long reads	FALCON	https://github.com/PacificBiosciences/FALCON	10.1038/nmeth.4035	2021
Low-accuracy long reads	Flye	https://github.com/fenderglass/Flye	10.1038/s41587-019-0072-8	2022
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Low-accuracy long reads	miniasm	https://github.com/lh3/miniasm	10.1038/nmeth.4432	2020
Low-accuracy long reads	NECAT	https://github.com/xiaochuanle/NECAT	10.1038/s41467-020-20236-7	2021
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Low-accuracy long reads	Ra	https://github.com/lbcb-sci/ra	10.1109/ISPA.2019.8868909	2020
Low-accuracy long reads	Raven	https://github.com/lbcb-sci/raven	10.1038/s43588-021-00073-4	2021
Low-accuracy long reads	SMARTdenovo	https://github.com/ruanjue/smartdenovo	10.20944/preprints202009.0207.v1	2021
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Low-accuracy long reads	wtdbg2	https://github.com/ruanjue/wtdbg2	10.1038/s41592-019-0669-3	2021
Low-accuracy long reads	shasta	https://github.com/paoloshasta/shasta	10.1038/s41587-020-0503-6	2023
High-accuracy long reads	Alice-asm	https://github.com/rolandfaure/alice-asm		2024
High-accuracy long reads	Flye	https://github.com/fenderglass/Flye	10.1038/s41587-019-0072-8	2022
High-accuracy long reads	HiCanu	https://github.com/marbl/canu	10.1101/gr.215087.116	2021
High-accuracy long reads	hifiasm	https://github.com/chhylp123/hifiasm	10.1038/s41592-020-01056-5	2022
High-accuracy long reads	IPA	https://github.com/PacificBiosciences/pbipa		2021
High-accuracy long reads	LJA	https://github.com/AntonBankevich/LJA	10.1101/2020.12.10.420448	2021
High-accuracy long reads	mdBG	https://github.com/ekimb/rust-mdbg/	10.1016/j.cels.2021.08.009	2021



Some terminology

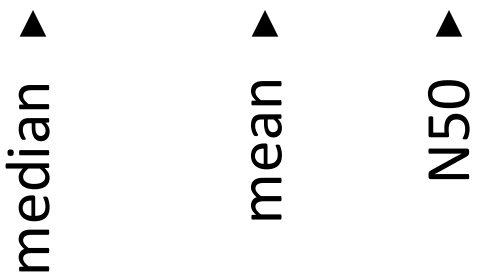
- Genome assembly methods do not always reconstruct full molecules (plasmids, chromosomes)
- Basic reconstructed fragments are contigs and scaffolds

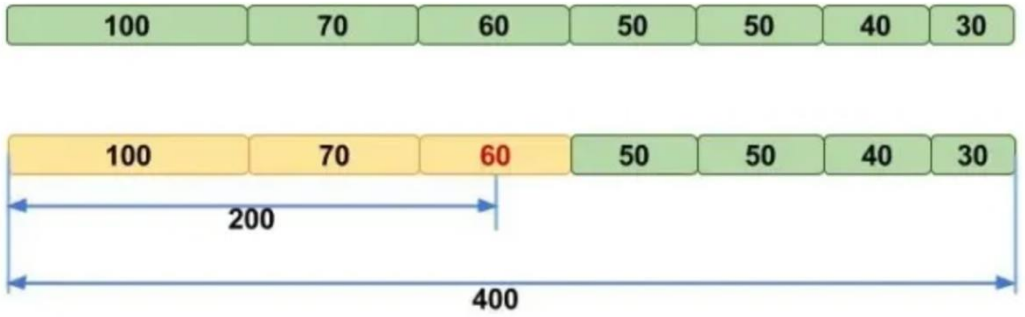
- *Contig*: contiguous sequence
- *Scaffold*: multiple ordered contigs, perhaps with gaps (NNNs)
- *Coverage*: sampling depth of the genome, typically >30×
- *N50*: typical contig/scaffold/fragment length



N50: typical fragment length

- Length-weighted median length
- “Completeness” of an assembly (50% of the genome is in contigs of size N50)
- Or:
  - sort all fragments by length
  - record the cumulative assembly length
  - the N50 is equal to the fragment length at which the cumulative length is 50% of the total length
- Higher N50: more complete and higher quality





Short-read assembly quality

Small, simple genomes



100% complete
1–100 contigs
N50 ~1 Mbp

Large, complex genomes





<90% complete
>10000 contigs
N50 ~100 kbp

Massive genomes





<50% complete
millions of contigs
N50 <1000 bp

The first problem: sequencing limitations



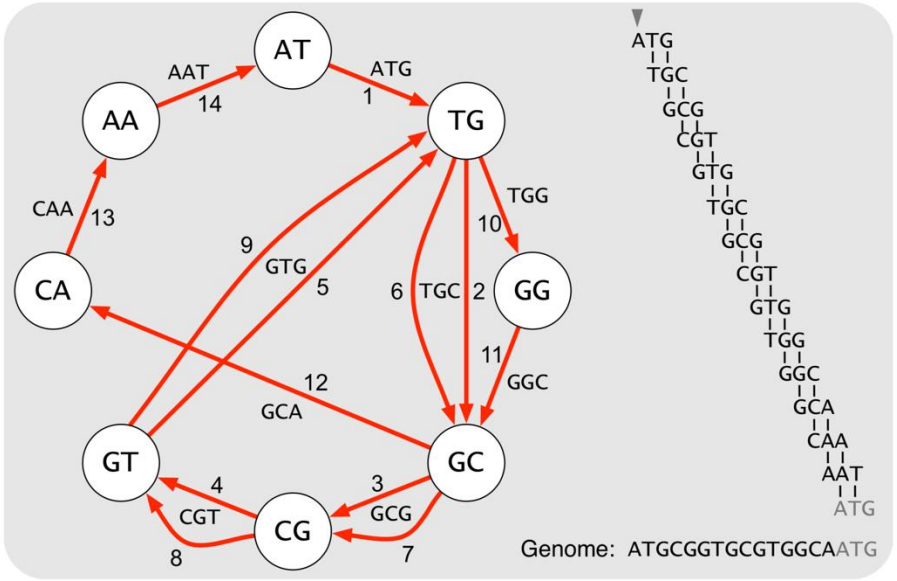
The second problem: technical errors

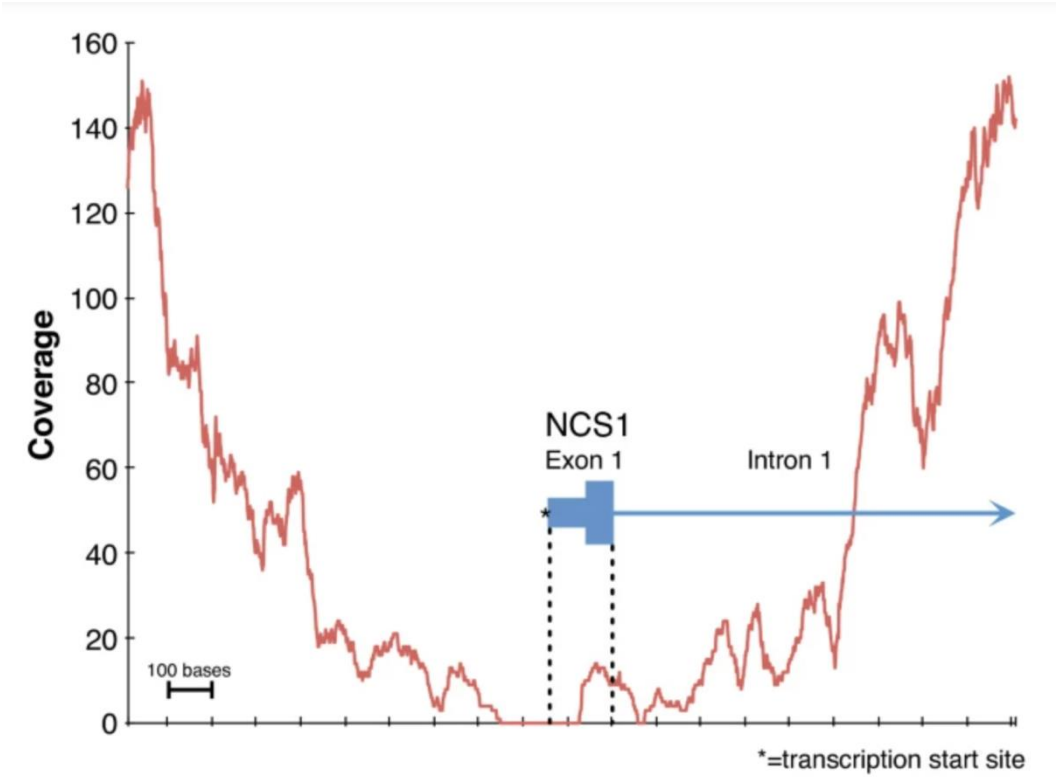


The third problem: genomic architecture

Technical challenges

- Uncorrected sequencing errors: misassemblies
- Sequencing bias: no or low coverage





"..all technologies exhibit error-rate biases in high- and low-GC regions resulting in low coverage"

Ross, M.G., Russ, C., Costello, M. *et al.* Characterizing and measuring bias in sequence data. *Genome Biol* **14**, R51 (2013). https://doi.org/10.1186/gb-2013-14-5-r51

The first problem: sequencing limitations

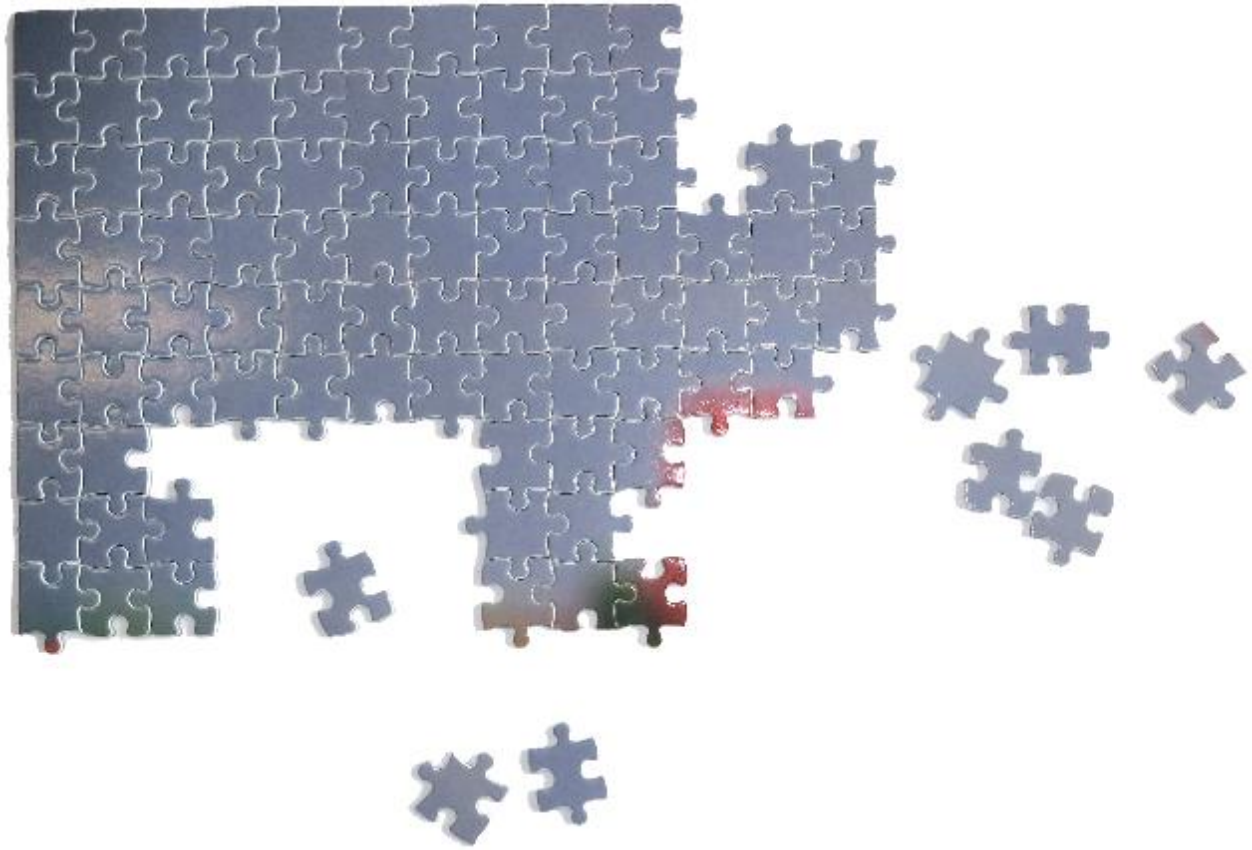


The second problem: technical errors



The third problem: genomic architecture

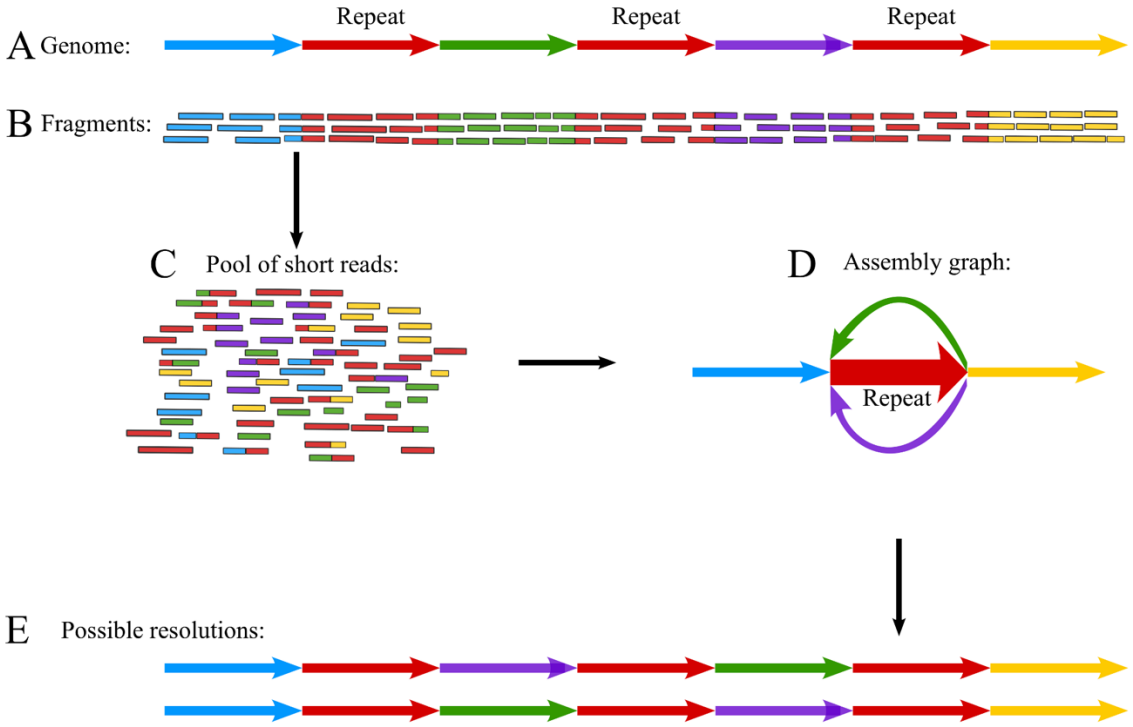
The third problem of genome assembly



Local genomic content may not contain enough information
to reconstruct the global structure

Repetitive content complicates assembly

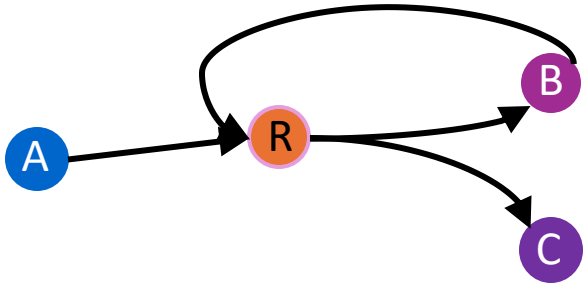
REPEAT	ANNOTATION
5′-AAAAAAAAAAAAAAA-3′	5′-$(A)_{15}$-3′
5′-ATATATATATATATAT-3′	5′-$(AT)_{8}$-3′
5′-GCCGCCGCCGCCGCCGCCGCC-3′	5′-$(GCC)_{8}$-3′
5′-GATCGATCGATCGATCGATCGATC-3′	5′-$(GATC)_{6}$-3′
5′-GCTCCGCTCCGCTCCGCTCCGCTCC-3′	5′-$(GCTCC)_{5}$-3′
5′-AAAATTAAAATTAAAATTAAAATT-3′	5′-$(AAAATT)_{4}$-3′



Repetitive content complicates assembly



genome



‘graph’ data structure



best path assembly



shortcut assemblies



safe contigs

Other sources of potential complications..

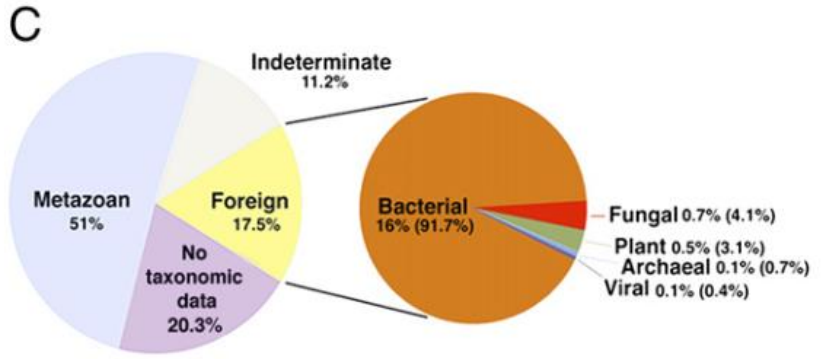
PNAS

Evidence for extensive horizontal gene transfer from the draft genome of a tardigrade

Thomas C. Boothby[a,1], Jennifer R. Tenlen[a,2], Frank W. Smith[a], Jeremy R. Wang[a,b], Kiera A. Patanella[a], Erin Osborne Nishimura[a], Sophia C. Tintori[a], Qing Li[c], Corbin D. Jones[a], Mark Yandell[c], David N. Messina[d], Jarret Glasscock[d], and Bob Goldstein[a]

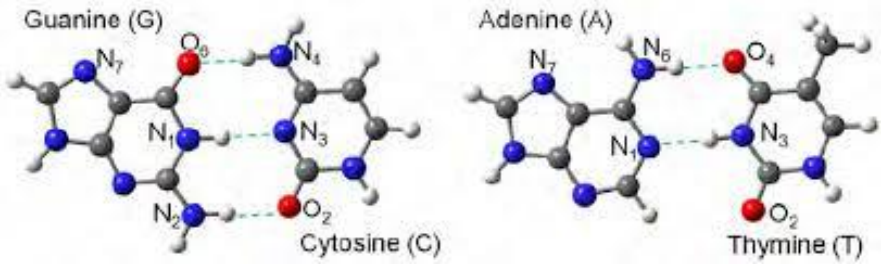
[a]Department of Biology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599; [b]Department of Genetics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599; [c]Eccles Institute of Human Genetics, University of Utah, Salt Lake City, UT 84112; and [d]Cofactor Genomics, St. Louis, MO 63110



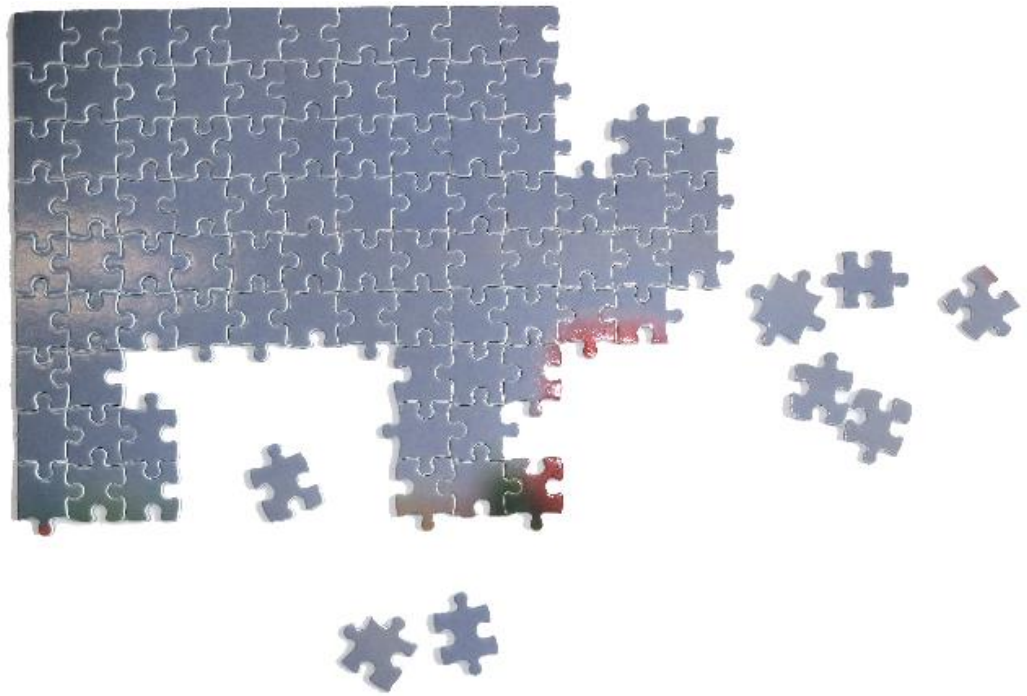


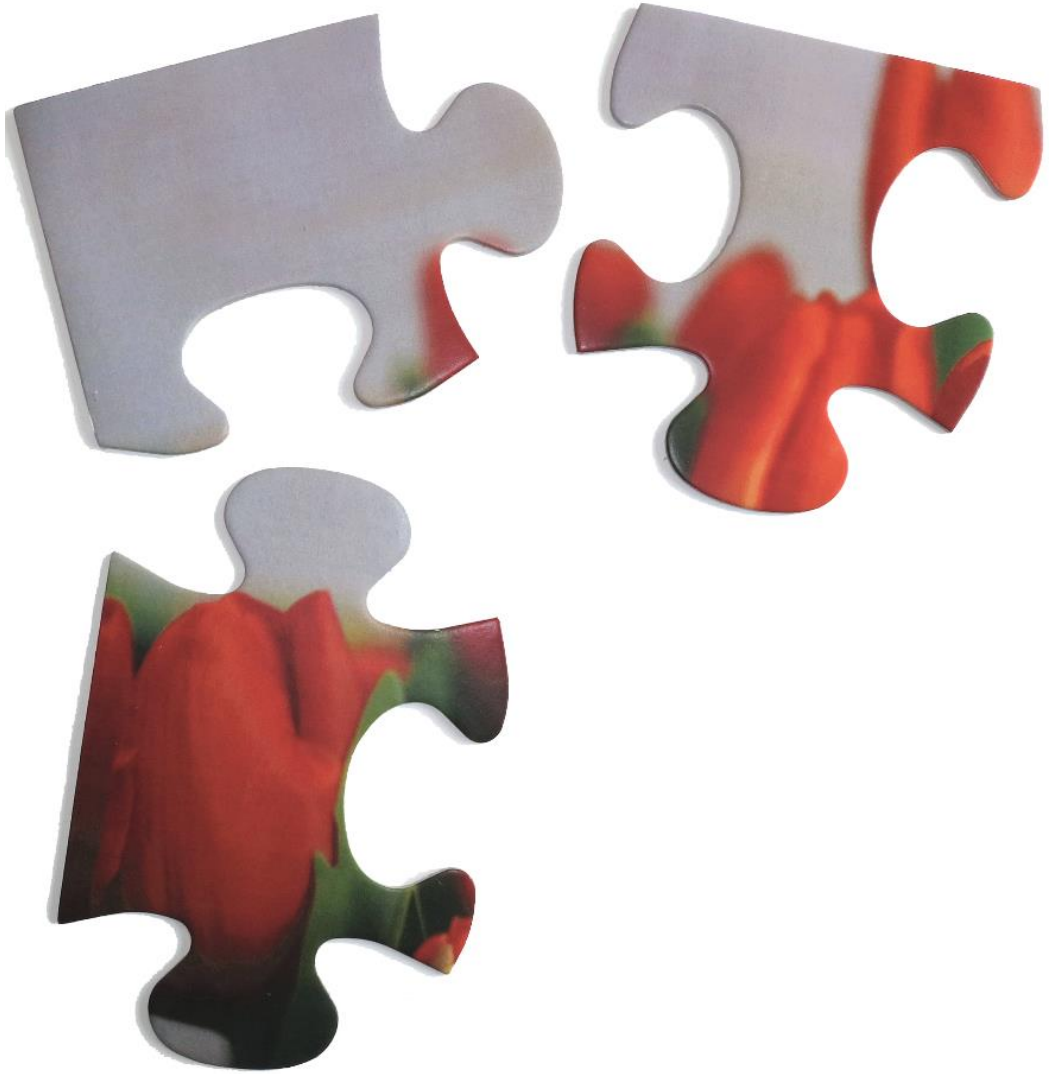
More assembly difficulties

- Chromosome ends
- Uncorrected sequencing errors
- Polymorphisms
- No or low coverage (sequencing bias)
- Repetitive structures
- Sequencing error
- Sequencing bias: or example, Illumina sequencing platform has base composition bias (favoring GC balanced regions), which usually results in uneven sequencing depth across genome;

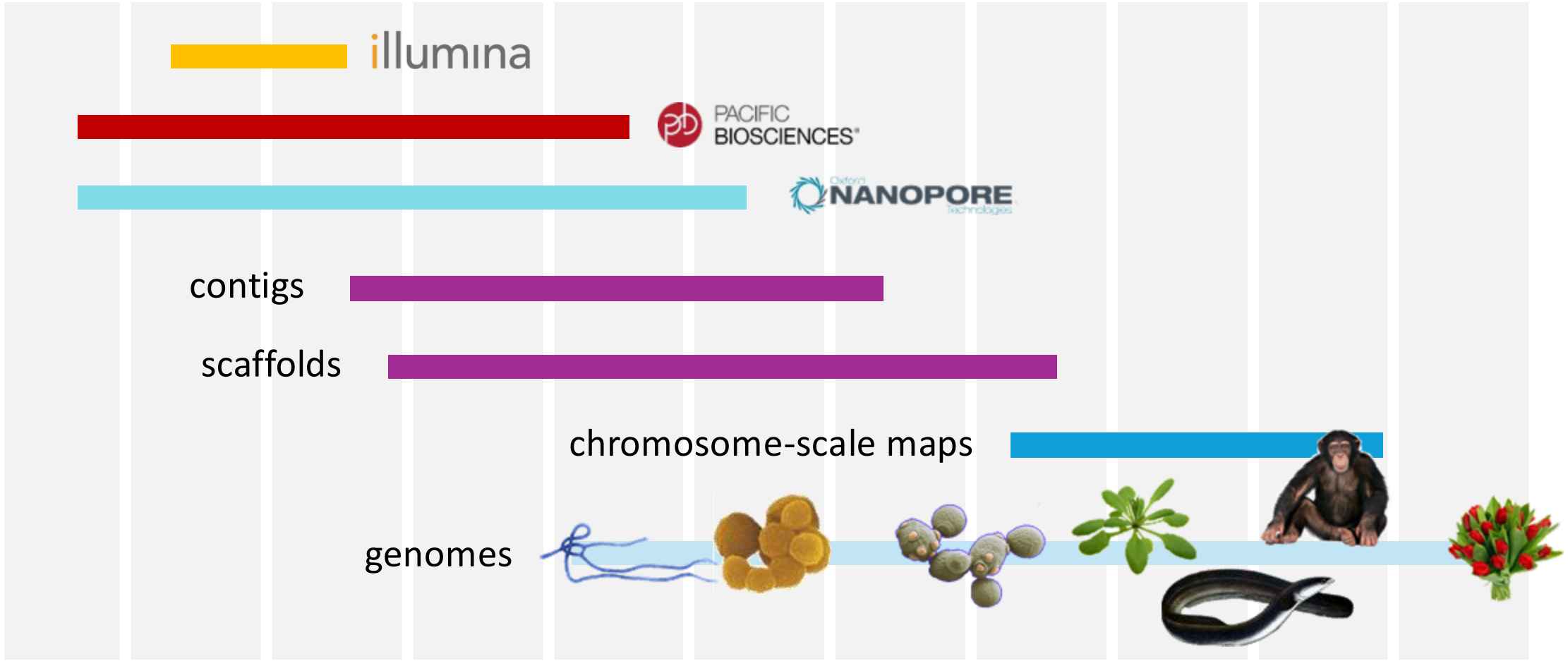


Assembly using long reads

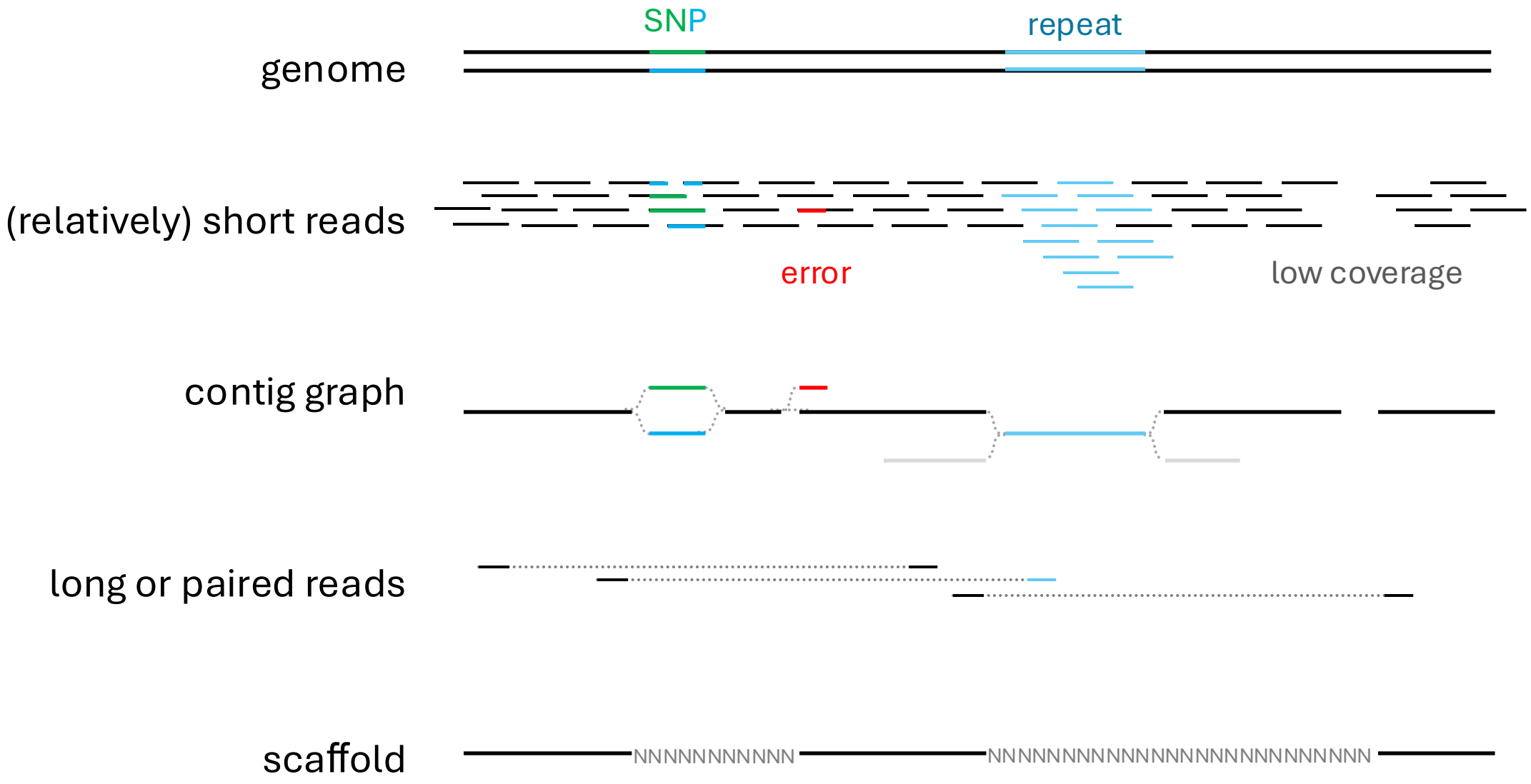




Genome sequencing



From genome to contigs and scaffolds



Assembly using long reads

- Current long read technologies provide sufficient amounts of reads to support *de novo* assembly (instead of scaffolding use only)
- This is useful, as long reads span ambiguities in the genome/graph (e.g. LTR retrotransposons: ~10 Kbp)

- Problem: the sequence identity is relatively low (raw ~98 %, HiFi/duplex better)
- Therefore it is harder to find reliable overlaps
- Complicated graph and high computational demands

- Sequences have to be **polished** by taking the consensus of high coverage → Q50 (<1 mistake per 100 Kbp)

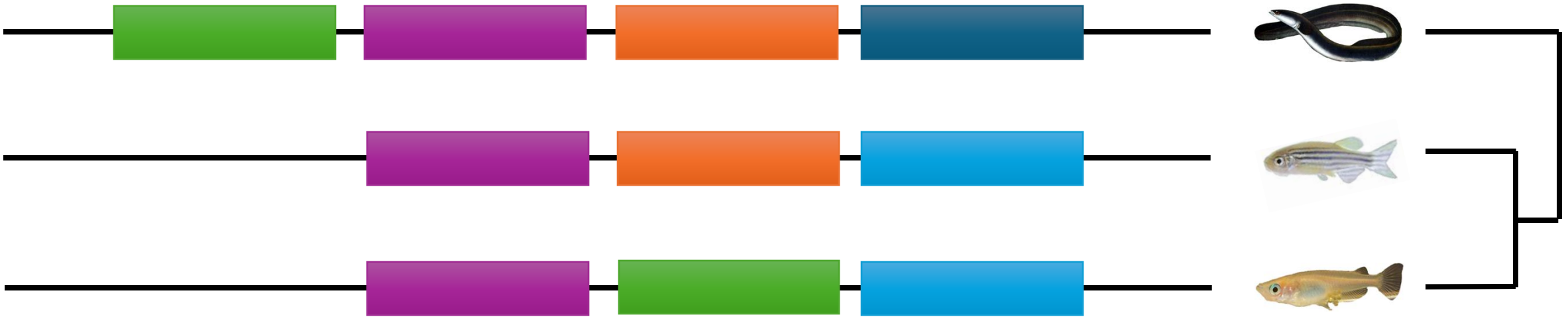
Our European eel genome assemblies

		Sequencing	Assembly	Statistics
I	*Evolution of complex body plans* (2010–2012)	Illumina GAIIx	CLC bi0 SSPACE	186281 scaffolds N50 78 kbp
II	*First vertebrate genome assembled using nanopore* (2015–2017)	Oxford Nanopore MinION R7–R9 Illumina HiSeq	TULIP	2366 scaffolds N50 1.23 Mbp
III	*Phylogenomics resolve early teleost evolution* (2019–2023)	Oxford Nanopore PromethION R9.4 Illumina HiC	wtdbg2 Juicebox	19 scaffolds N50 55.5 Mbp

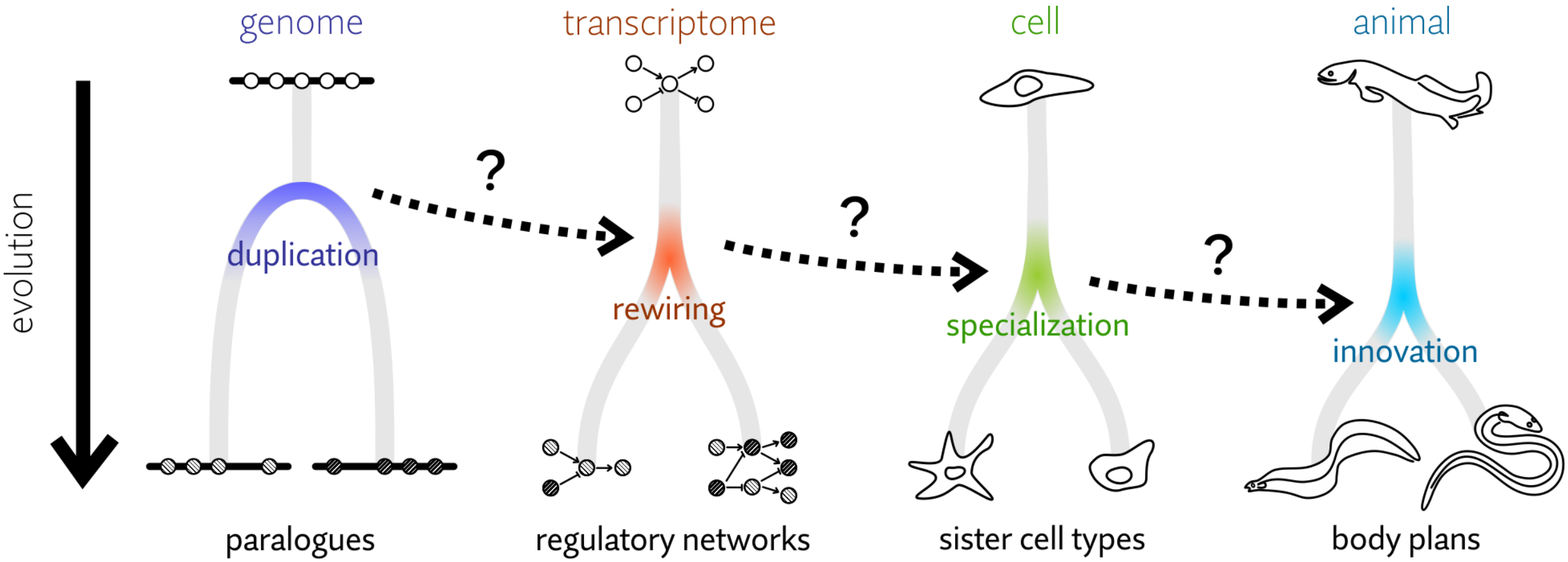


Long reads: Synteny analysis

- Synteny is the conservation of (local) gene order
- Offers a strong phylogenetic signal
- (Mostly) independent from sequence comparisons



From genome duplication to innovation?



Summary



Genome assembly from sequencing reads is a hard problem



All genome assemblies contain errors! Beware of contig ends and scaffold gaps, they may indicate problematic genomic complexity



With long-read sequencing technology, new bioinformatics, and mapping technologies, assemblies start to reach telomere-to-telomere quality

Why DNA sequencing?



structural/comparative genomics
genome sequencing



functional genomics
gene expression analyses, chromatin structure, DNA binding elements

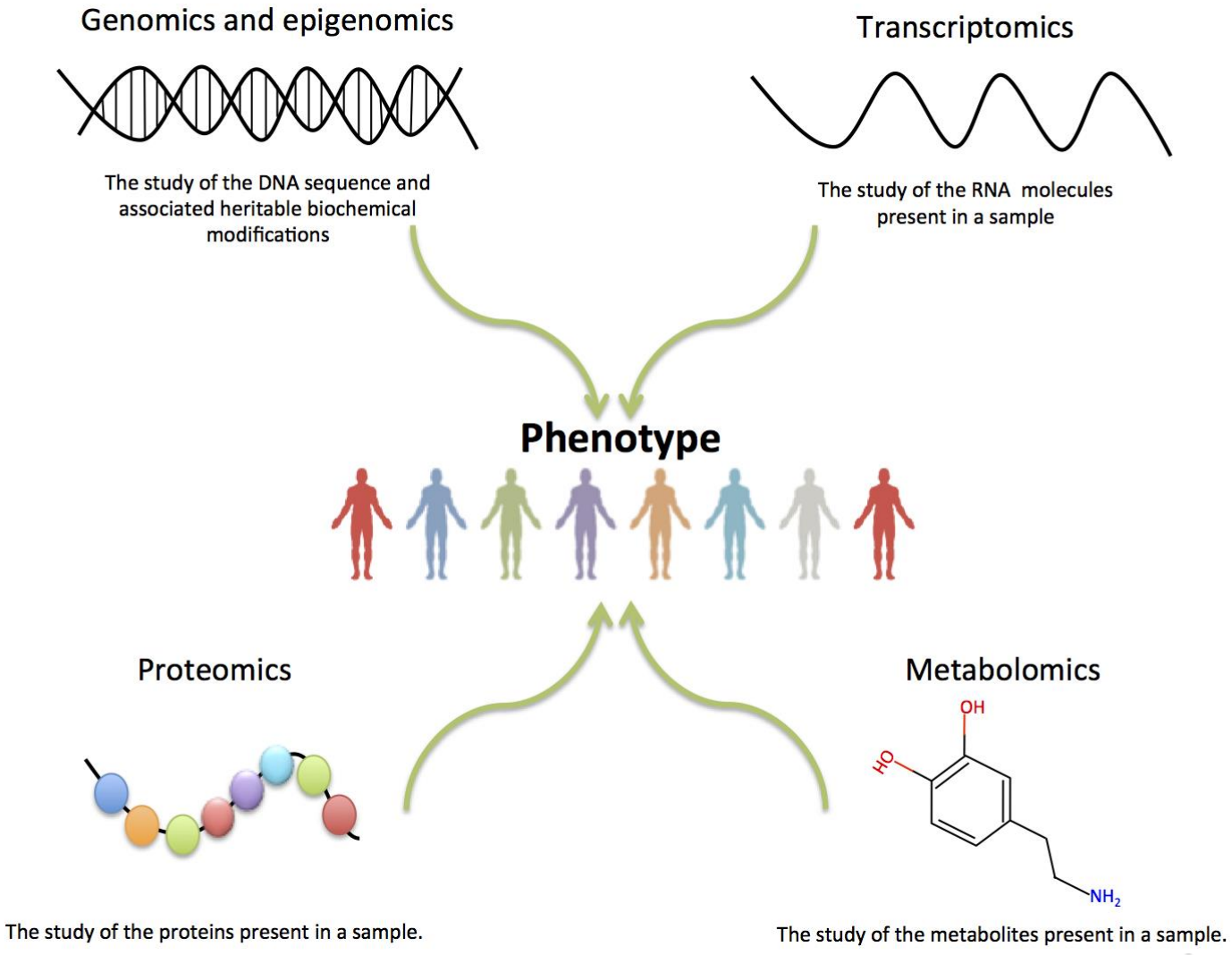


- ChIP-seq
  chromatin immunoprecipitation
- ATAC-seq
  assay transposase-accessible chromatin
- RNA-seq
  cDNA sequencing
- Single-cell RNA-seq
- Spatial transcriptomics



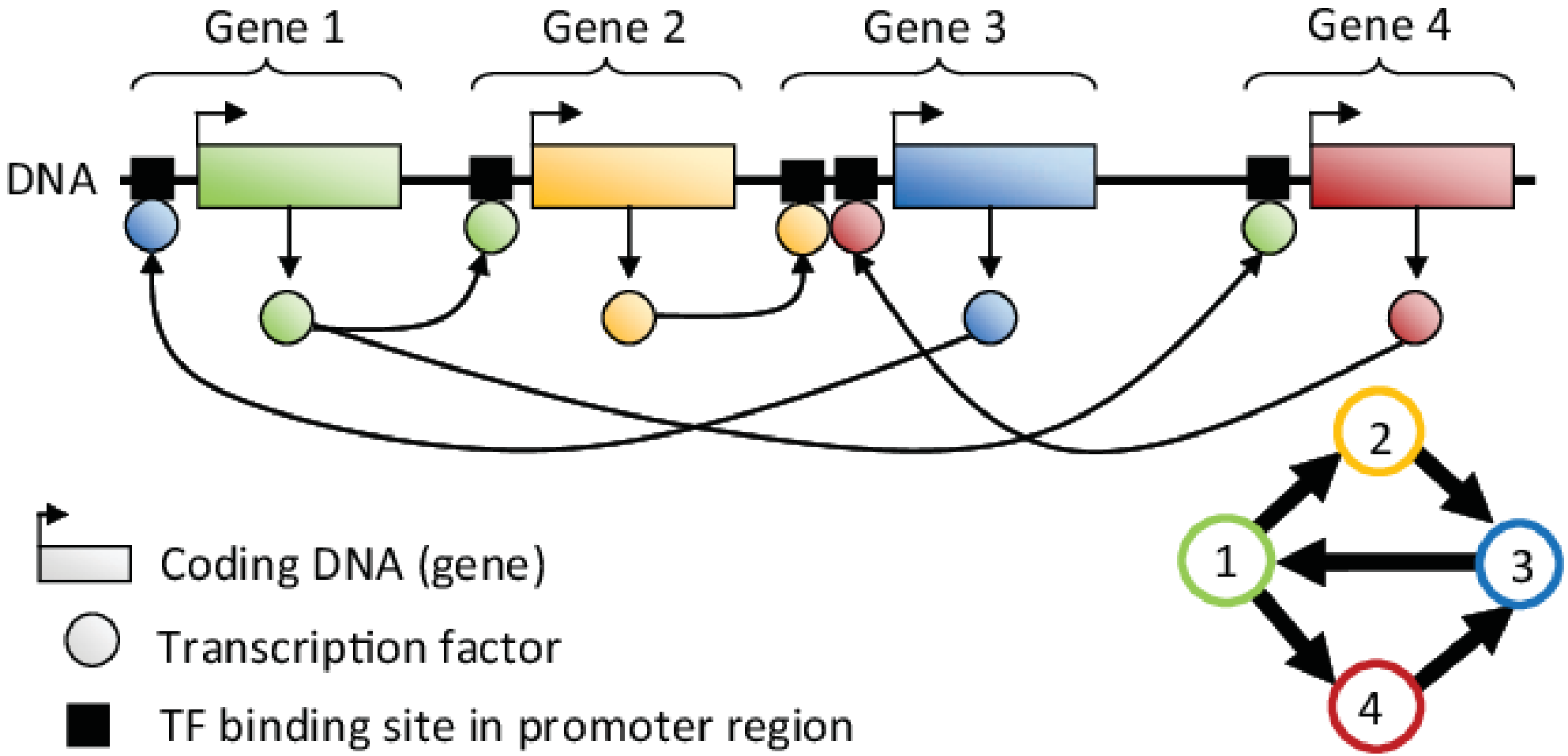
What is functional genomics?

- The field that aims to understand how genes lead to phenotypes
- Focuses on cellular information processing mechanisms
- Gene expression (transcriptomics)
- Regulation of gene expression (epigenetics)



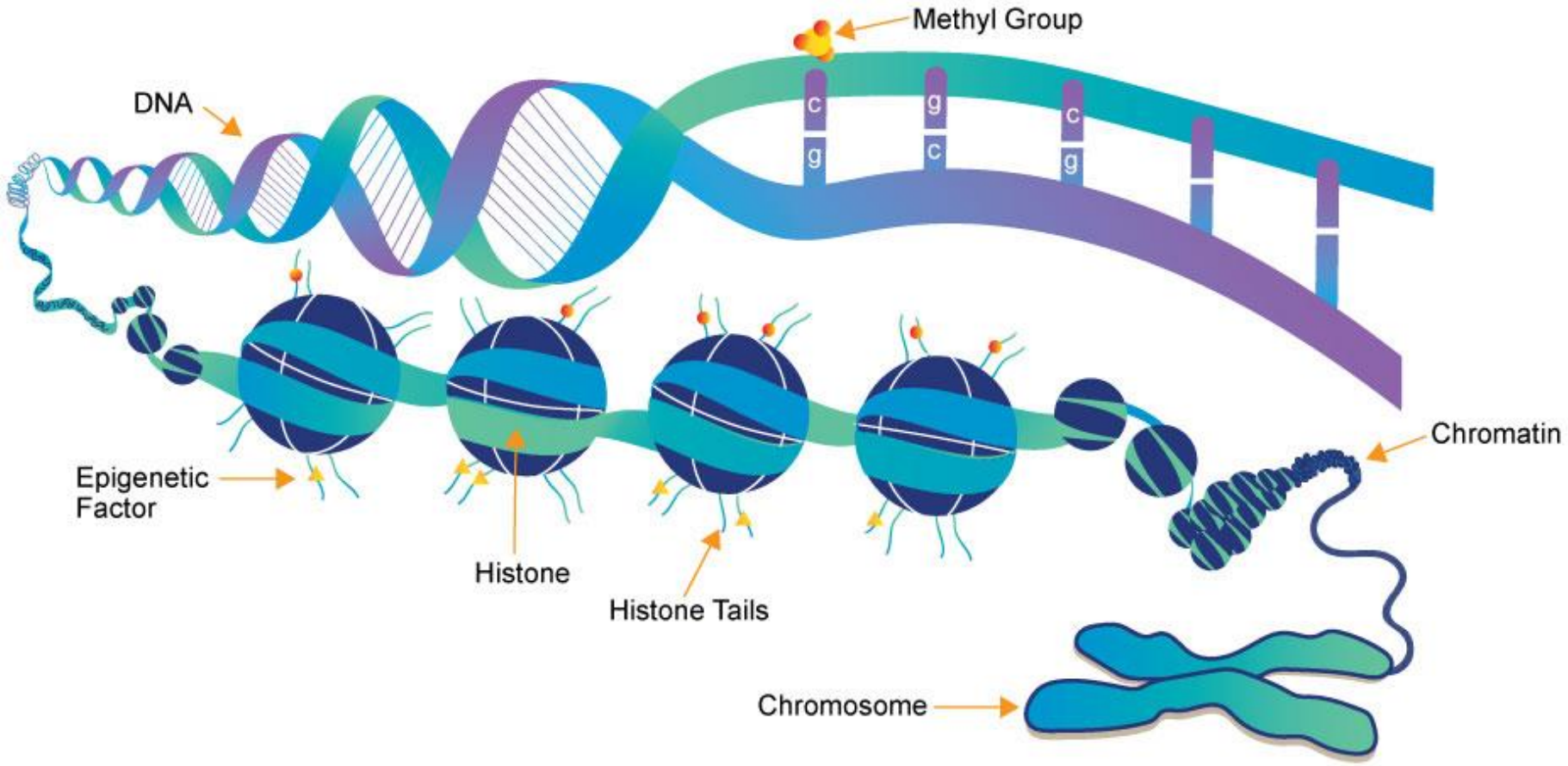
EMBL-EBI

Gene regulatory networks



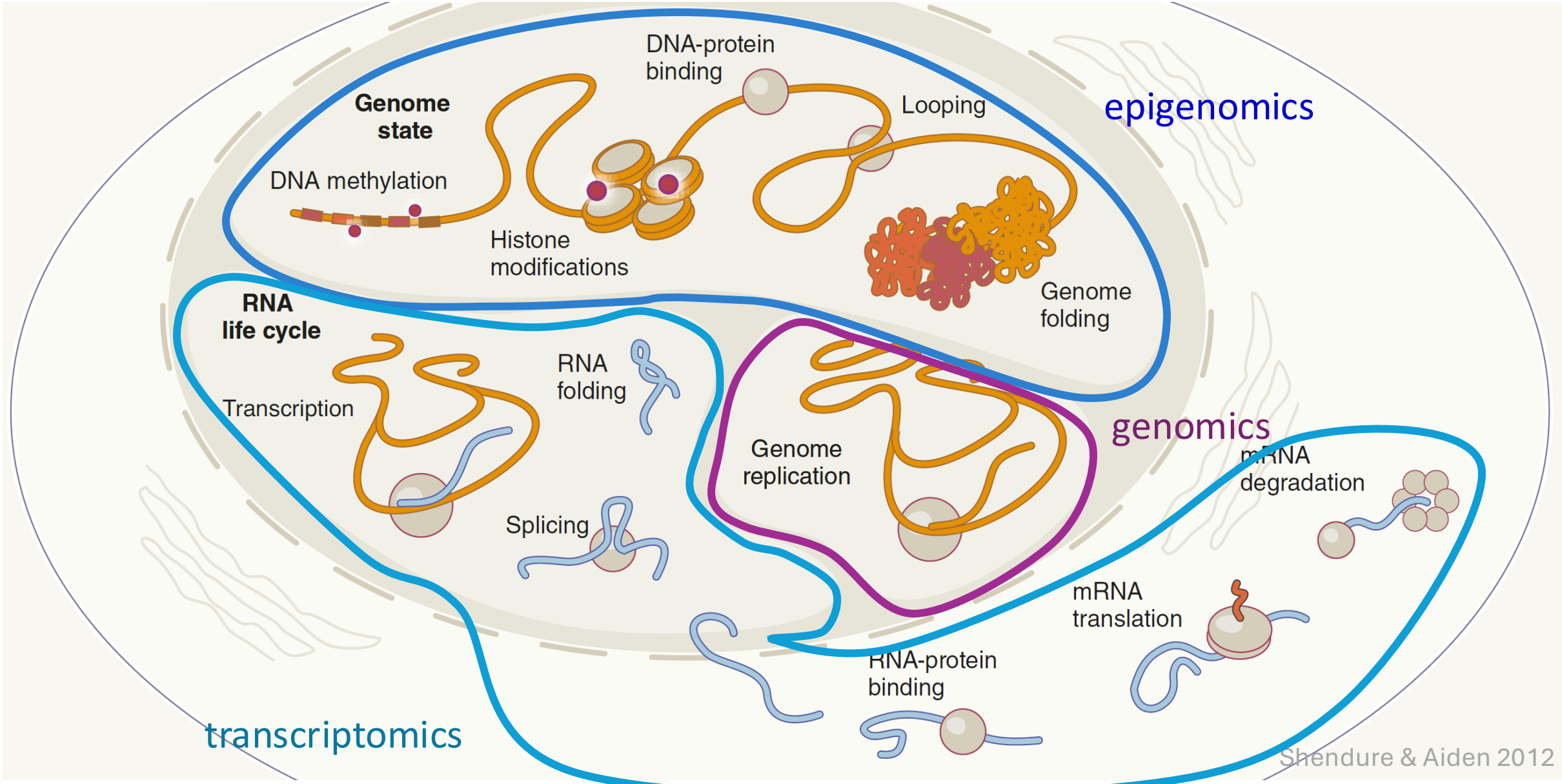
Rau *et al.* 2012

Epigenetics

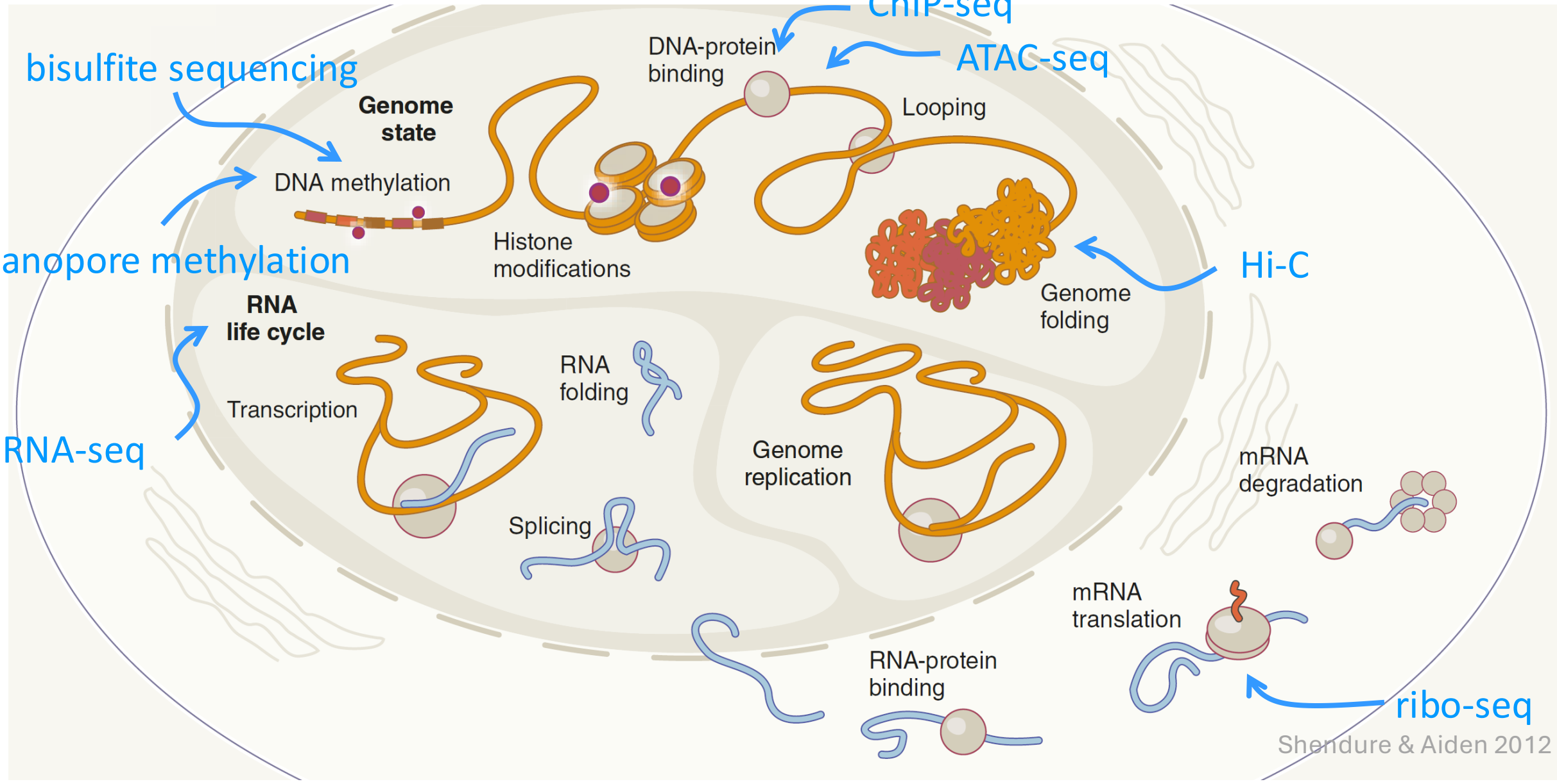


Epigenetics: non-hereditary changes to the genome regulate its activity

Cellular mechanisms



Cellular mechanisms



Shendure & Aiden 2012

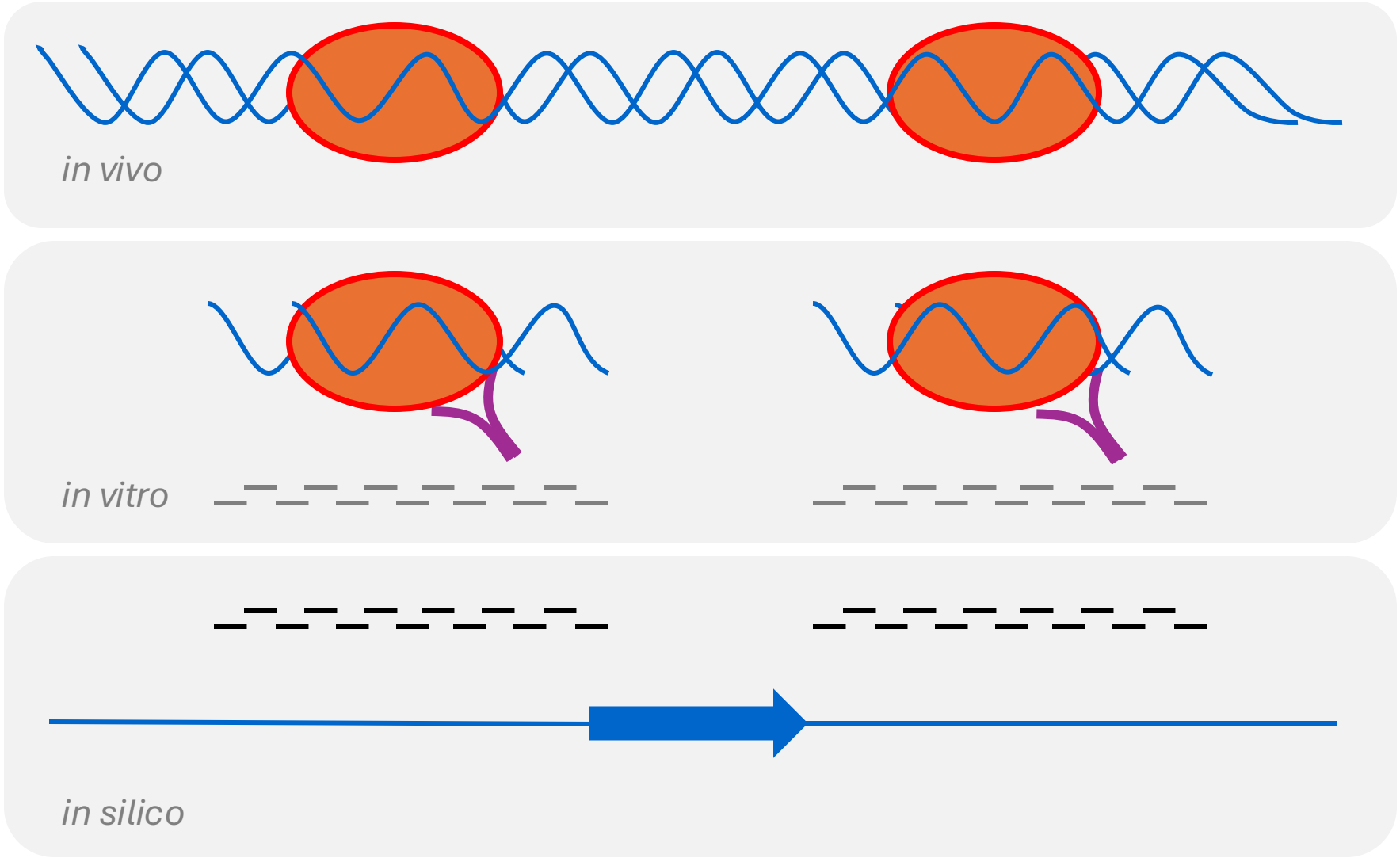
Major functional genomics protocols

- **ChIP-seq**: chromatin immunoprecipitation + sequencing
  Library of DNA binding sites to which *specific* proteins are bound (e.g. histones, transcription factors)
  Epigenomics assay for genome (in)activity

- **ATAC-seq**: assay for transposase-accessible chromatin + sequencing
  Library of DNA to which few/no proteins are bound
  Epigenomics assay for genome activity

- **RNA-seq:** transcriptome (mRNA) sequencing
  Library of cDNA
  Assay for transcriptional activity

ChIP-seq (immunoprecipitation)



(epi)genome
(DNA + protein)

crosslinking proteins,
immunoprecipitation

library of binding sites

reads

alignment

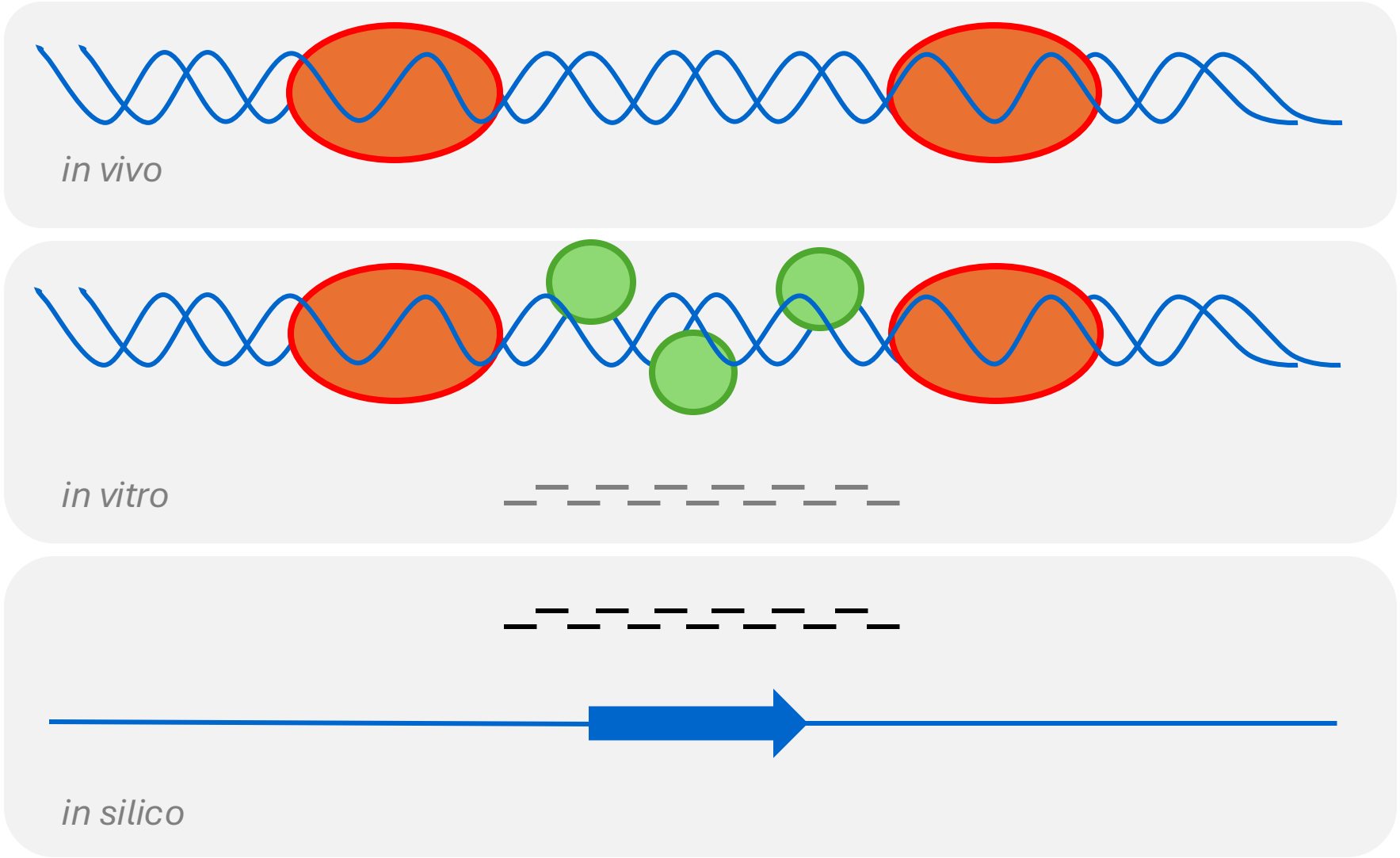
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  Library of cDNA
  Assay for transcriptional activity

ATAC-seq (transposase accessibility)



(epi)genome

‘tagmentation’

library

reads

alignment

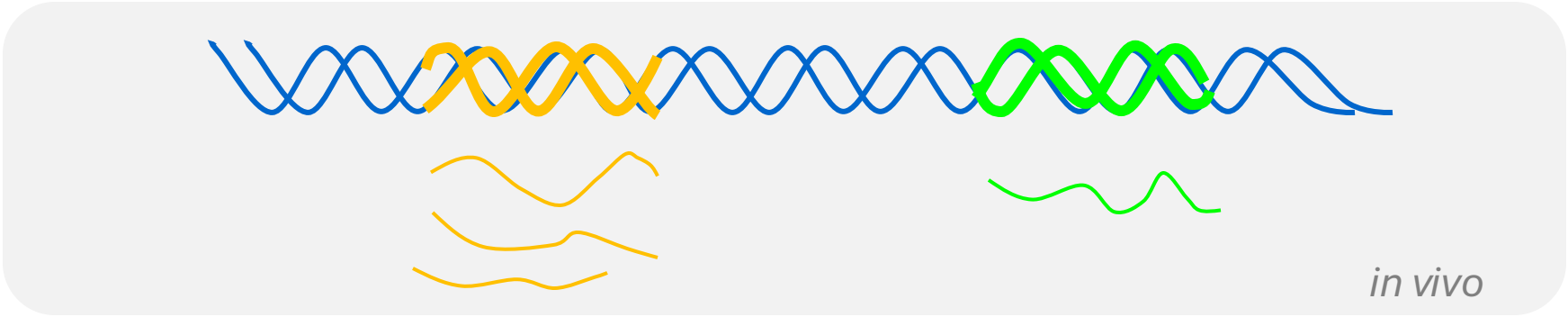
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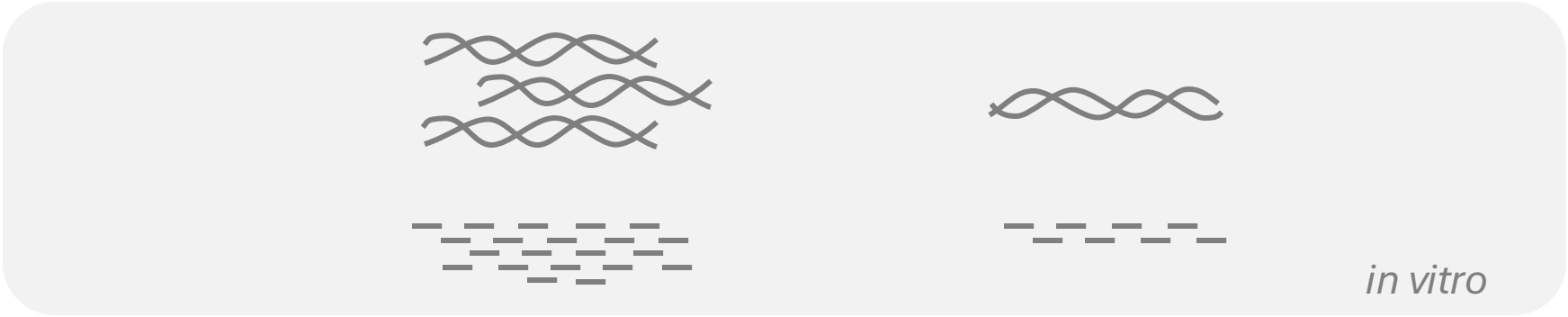
- **RNA-seq:** transcriptome (mRNA) sequencing
  Library of cDNA
  Assay for transcriptional activity

cDNA sequencing (RNA-seq)



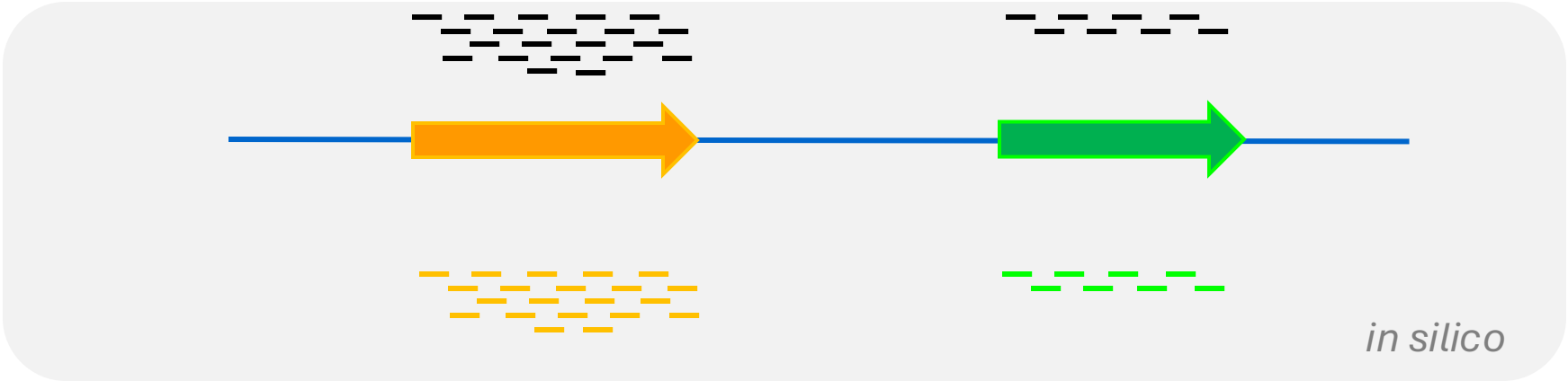
genome

mRNA



cDNA

library

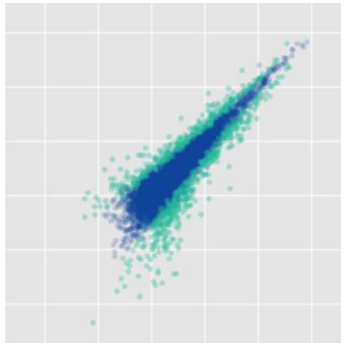


reads

reference

expression values

Transcriptomics applications



RNA-seq:

Precise expression levels for entire tissues

Comparative expression studies

Transcriptional regulation inference

Single-cell RNA-seq:

Single-cell library preparation for sequencing

Cell type discovery

Spatial transcriptomics

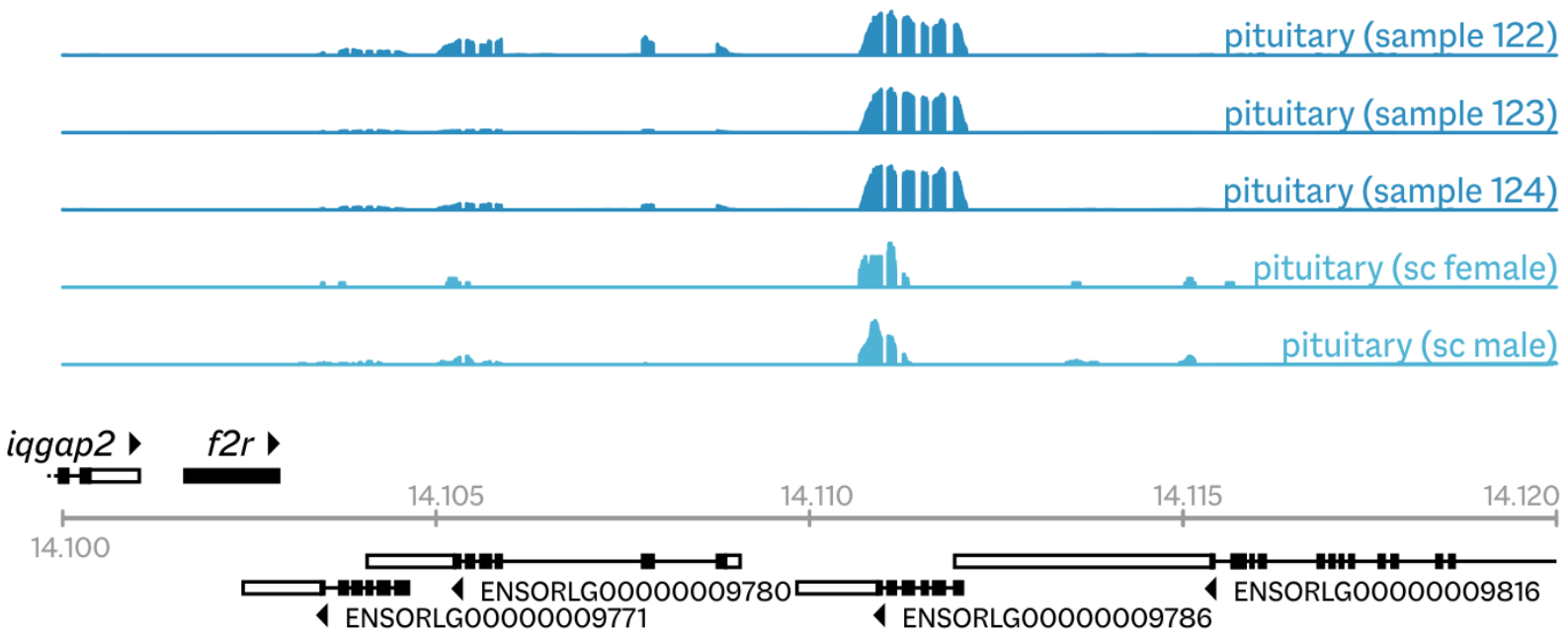
Lineage tracing

RNA-seq (transcriptomics)

- Deep sequencing of cDNA (exception: nanopore direct RNA)
- Uses a sequencer as a ‘molecule counter’
- Gene expression level ~ number of reads per gene
- Expression levels differ by several orders of magnitude
- Possible objectives:
  - quantification of gene expression (Illumina sequencing)
  - determination of differential gene expression
  - identification of splice isoforms (long-read sequencing)

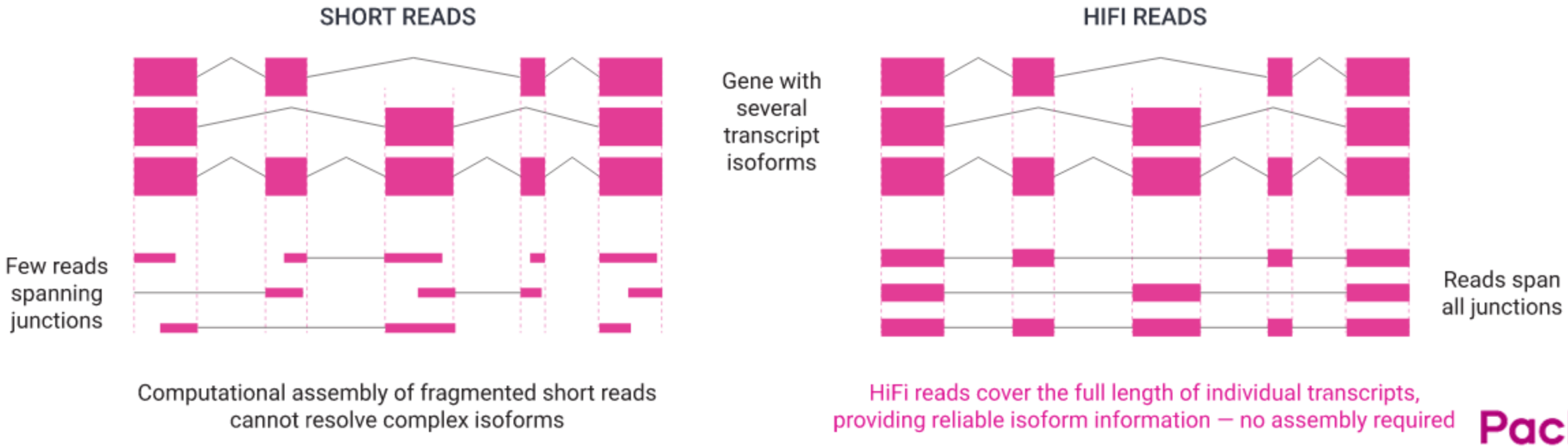
Short-read sequencing for RNA-seq

- Almost all RNA-seq uses Illumina short-read sequencing for molecule counting
- The number of reads aligning to a gene in the genome ~ the expression level
- Corrections necessary for gene length!

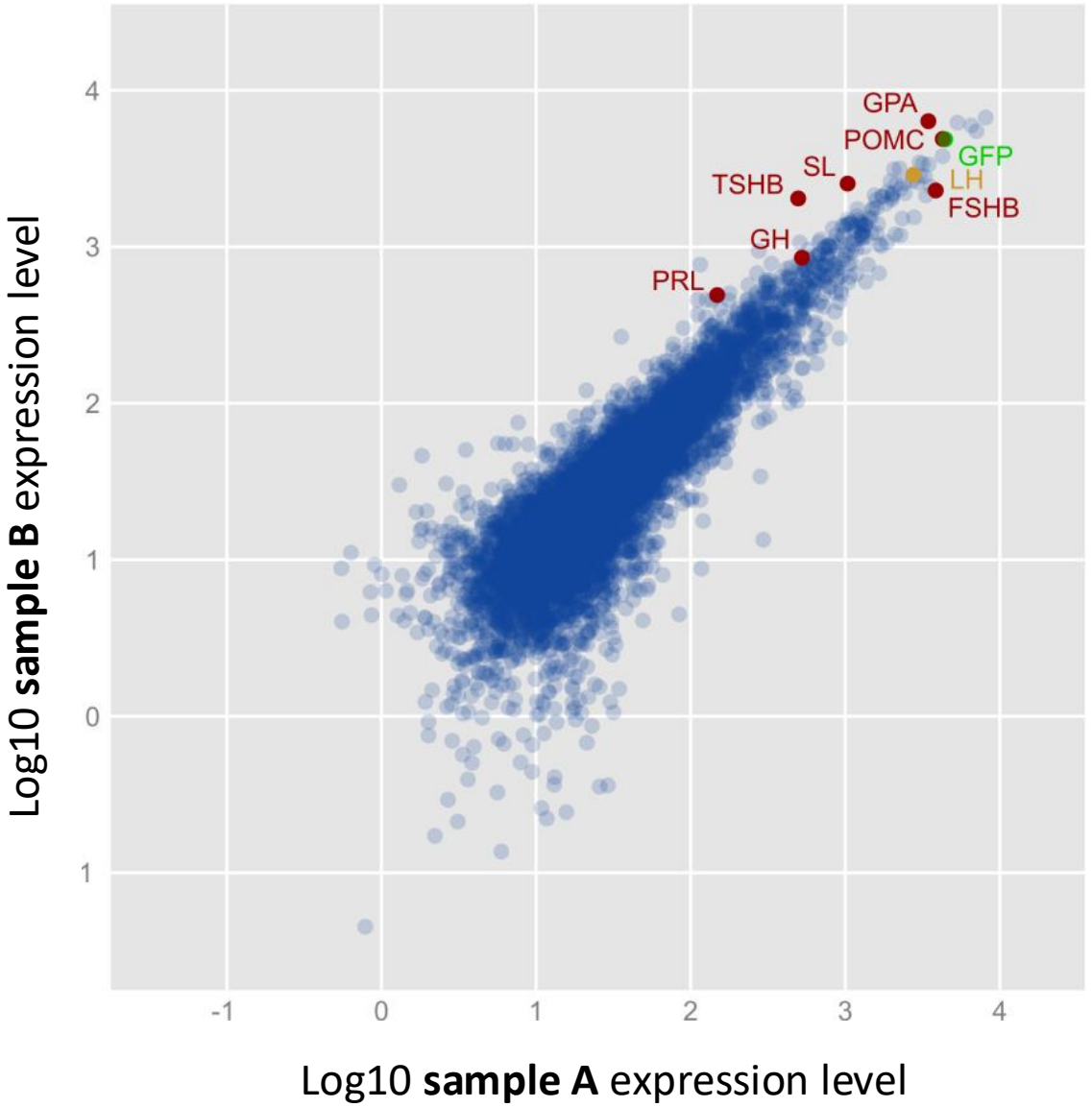


Long-read transcript sequencing

- Emerging as an alternative for short reads as 3rd-gen throughput increases
- Long-read sequencing potentially contains information on
  - Pre-mRNA Isoforms
  - Isoforms (splicing variants)

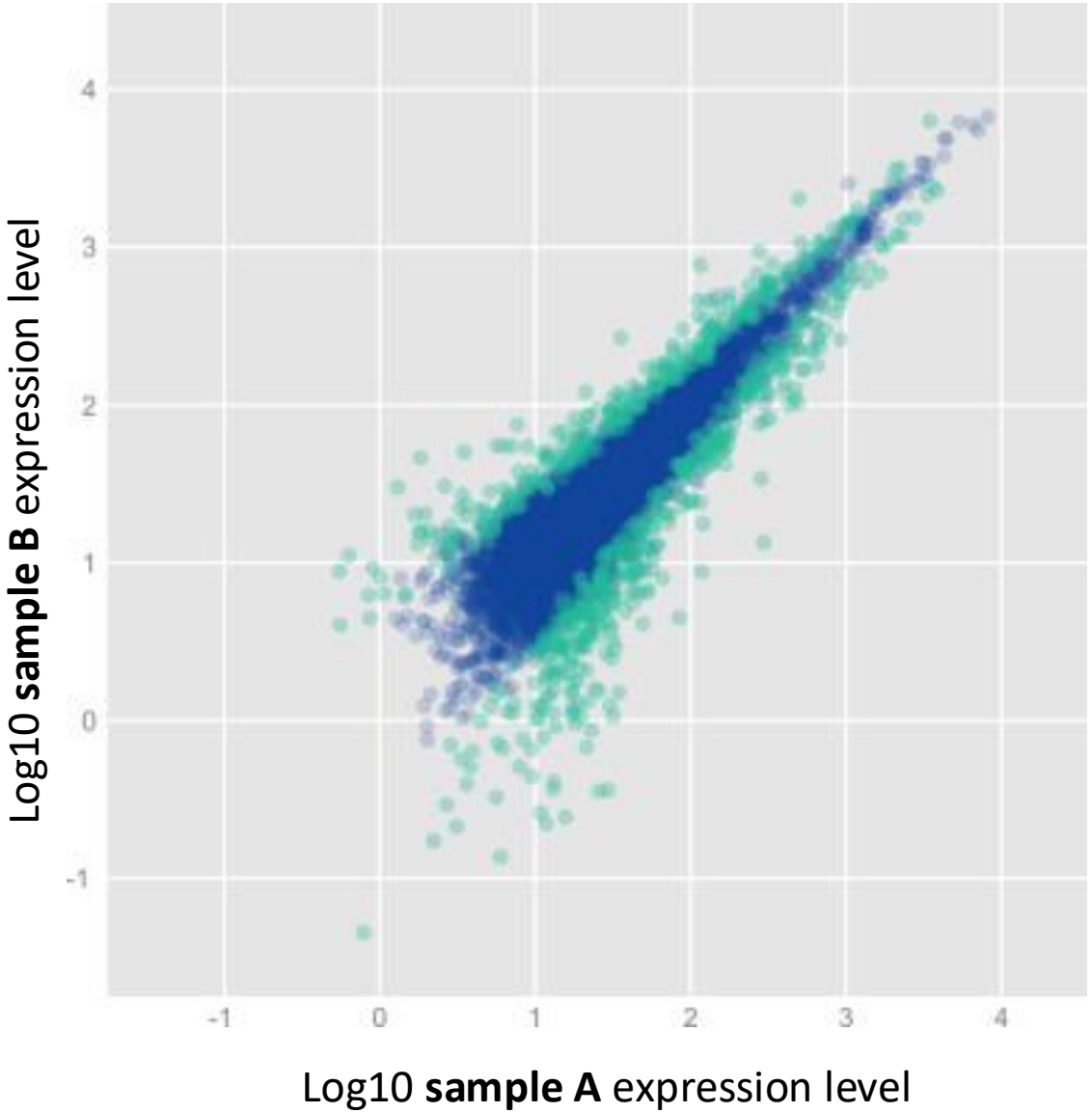


Differential gene expression



- 10–30 million reads/sample
- Every (blue) dot is a gene
- Note the logarithmic scale!

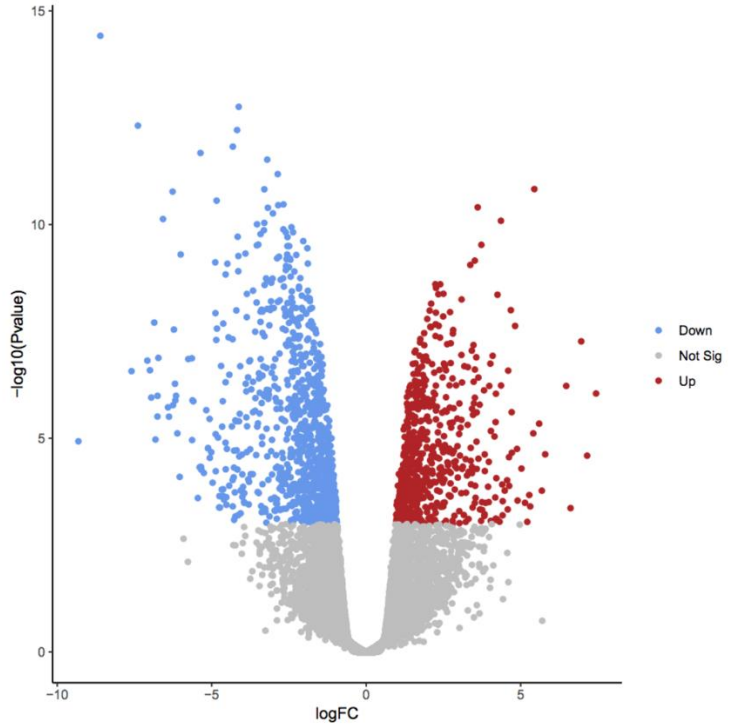
Differential gene expression

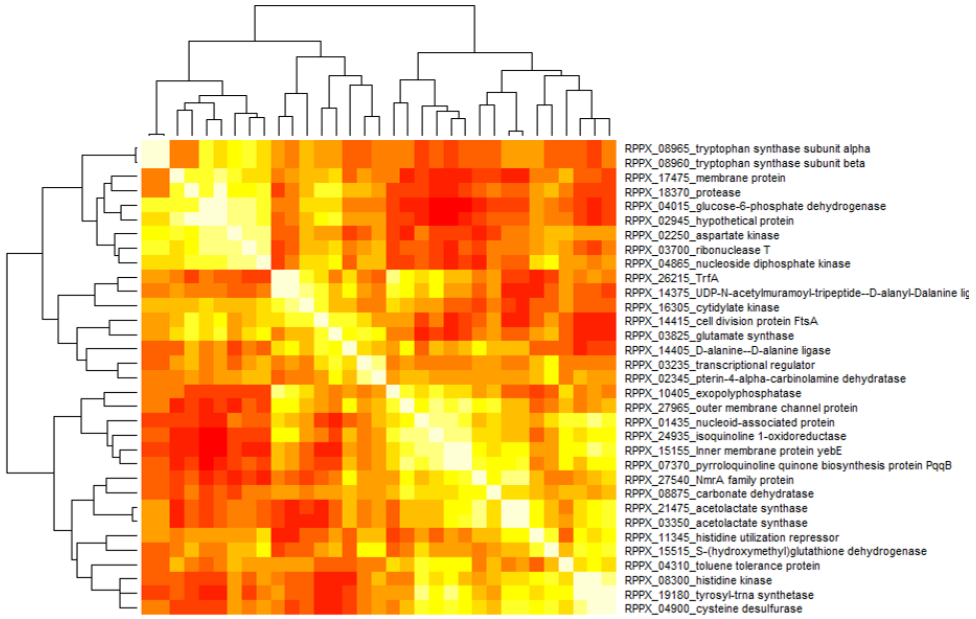


- 10–30 million reads/sample
- Every (blue) dot is a gene
- Note the logarithmic scale!
- Green: differentially expressed between the two conditions

Differential gene expression

Control vs. treatment





- Statistical assessment of which genes in expression level differ between samples
- Requires multiple (at least 3) samples per condition
- Only meaningful for intrinsically comparable samples!
- Normalization: making genes and samples comparable (by e.g. scaling)

Transcriptomics applications

RNA-seq

Precise expression levels for entire tissues

Comparative expression studies

Transcriptional regulation inference



Single-cell RNA-seq:

Single-cell library preparation for sequencing

Cell type discovery

Spatial transcriptomics

Lineage tracing

Methods of the year

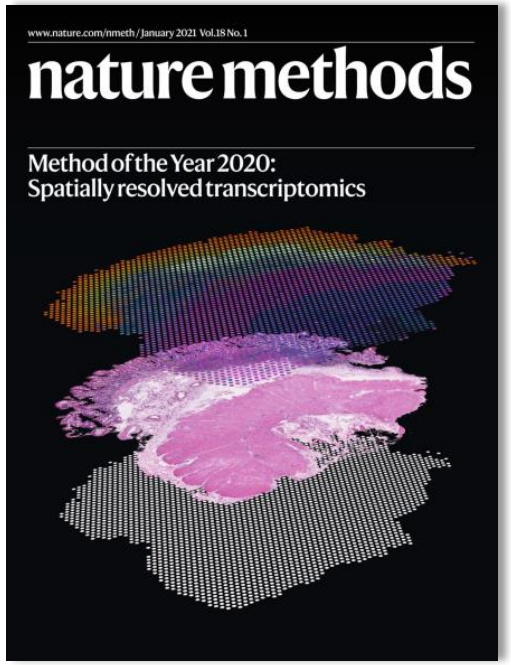
2013: single-cell sequencing

2019: single-cell multimodal omics analysis

2020: spatially resolved transcriptomics







Bulk



Bulk transcriptomic analyses lose spatial and single cell information

vs

Single cell



Single cell transcriptomic analyses retain single cell information but lose spatial organization

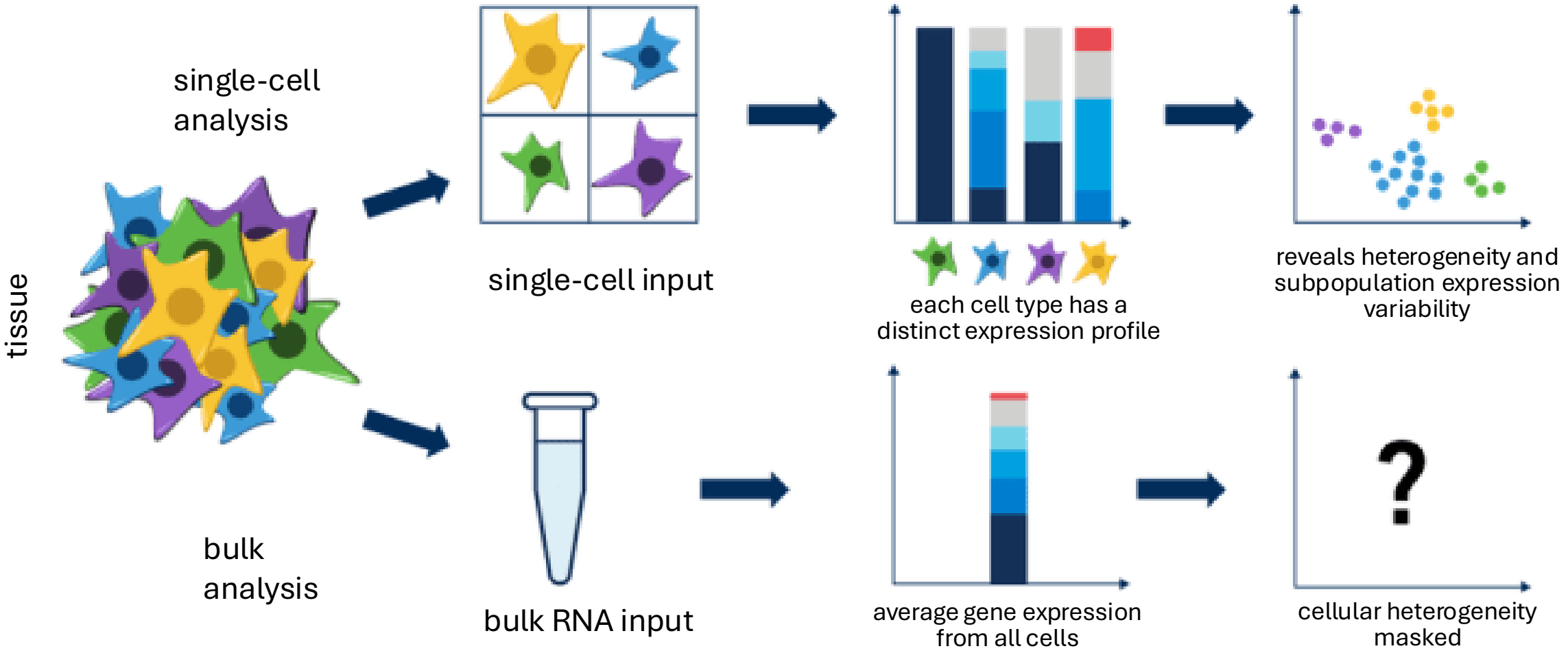
vs

Single cell spatial analysis



Single cell, multiplexing spatial analyses provide transcriptomic information with spatial organization while retaining single cell resolution

Single-cell and bulk RNA-seq



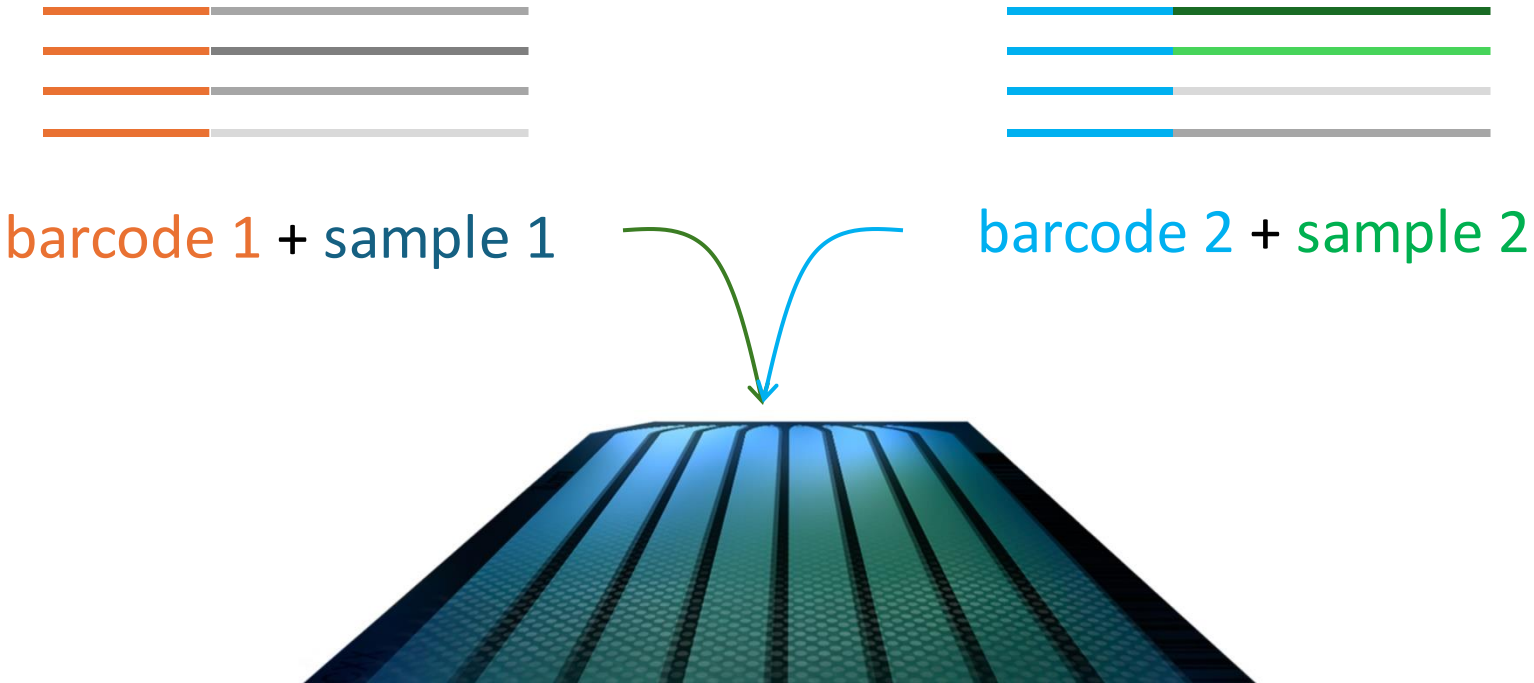
10x GENOMICS

Single-cell and bulk RNA-seq

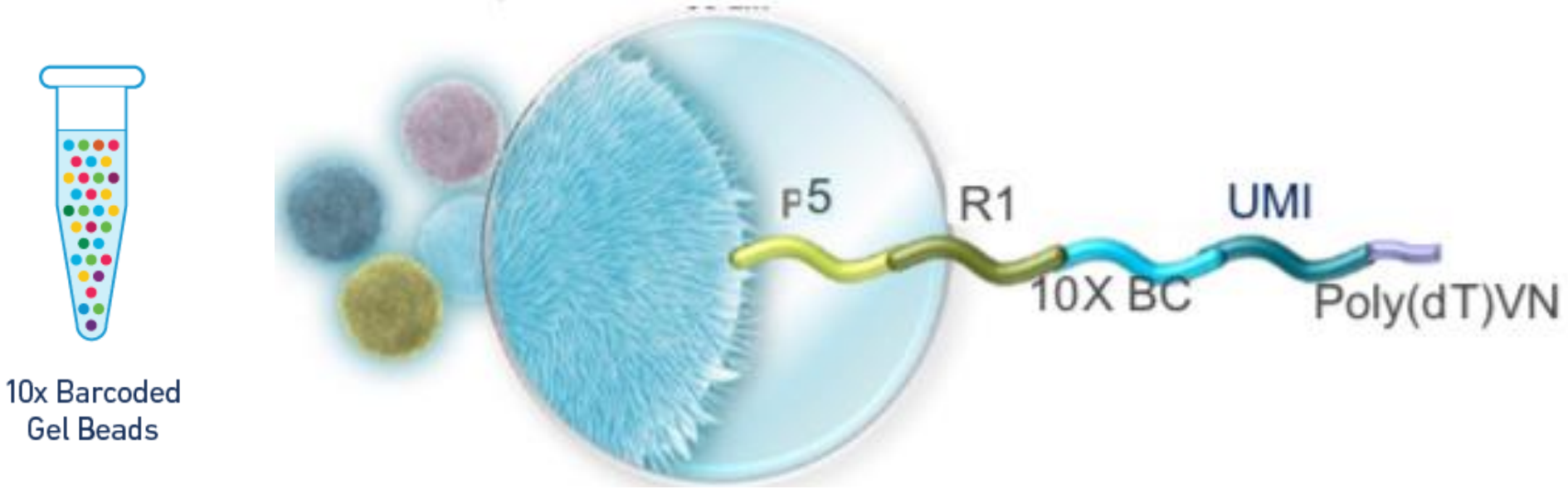
- Bulk RNA-seq: reliable transcript quantification, but averaged over all cells (cell types)
- Single-cell RNA-seq: less reliable quantification, but per cell type
  Artefacts caused by
  - low sequencing depth
  - 3′-biased protocols (polyA selection)
  - RT dropout (not all transcripts are amplified)
  - Computational processing
- Bulk RNA-seq is relatively affordable (but needs multiple samples),
  scRNA-seq is expensive (but you can get away with fewer samples)
- *Deconvolution* uses scRNA-seq profiles to interpret bulk RNA-seq samples

Illumina multiplexing (barcoding)

- You do not always need >200 million sequencing reads per sample
- RNA-seq depth: 10–50 million reads
- Solution: include unique ‘barcode’ sequences in the adapters
- This allows you to divide an Illumina flowcell lane amongst up to 380 samples



Single-cell RNA-seq



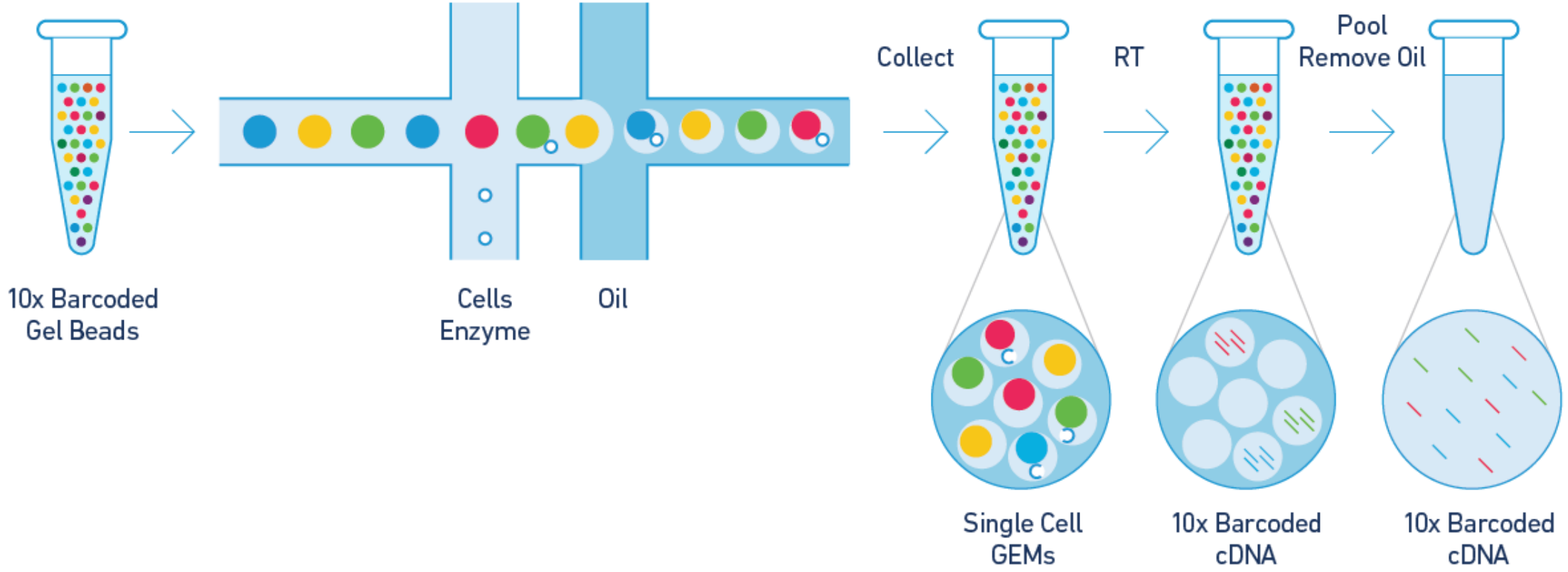
Two barcodes per cDNA:

- One cell-specific (gel bead-specific)
- One transcript-specific (universal molecular identifier UMI)

to enable precise tracing of reads to cells and transcripts.

10x GENOMICS

Single-cell RNA-seq



10x GENOMICS

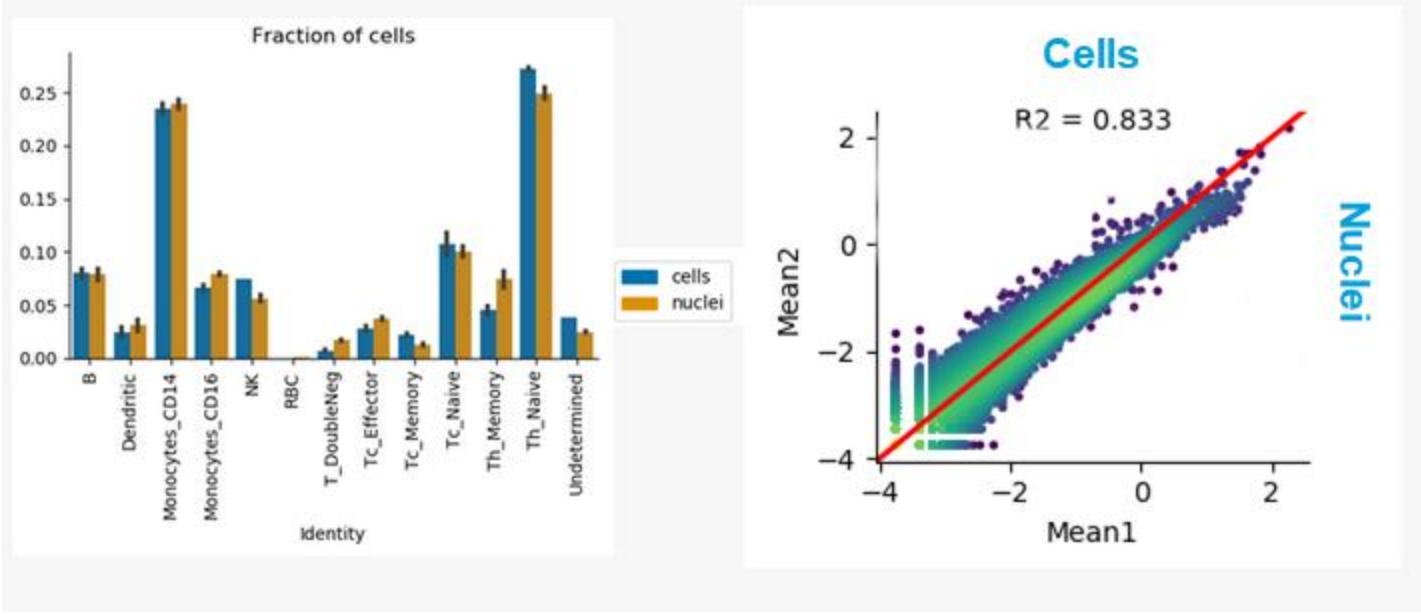
Single-cell RNA-seq



10x GENOMICS

Single-nuclei sequencing

- scRNA-seq requires dissociation of cells from a tissue (enzymatic, mechanic)
- This can damage cells (depending on the cell type)
  and lead to an altered representation of cellular abundance
- Alternative: use cellular nuclei instead

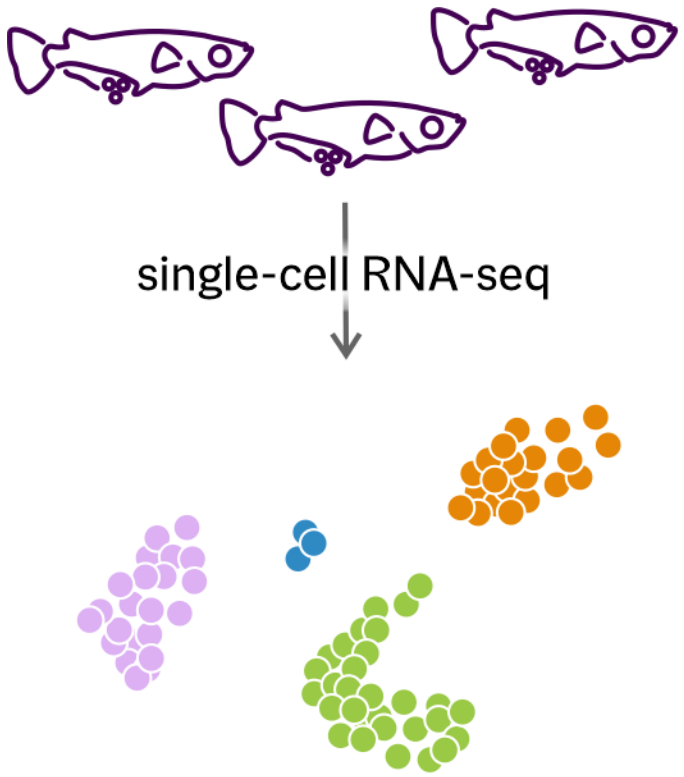


After sequencing: a large, sparse table

- 1000–1000000 cells
- 20000 genes
- Most entries are zero:
  - Sequencing depth is low-moderate (~50000 reads)
  - Not all transcripts are amplified (random dropout)
- Other issues:
  - Normalization
  - Doublets

	Cell1	Cell2	…	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
…	.	.	.	.
…	.	.	.	.
…	.	.	.	.
GeneM	25	0	.	0

Pituitary scRNA-seq

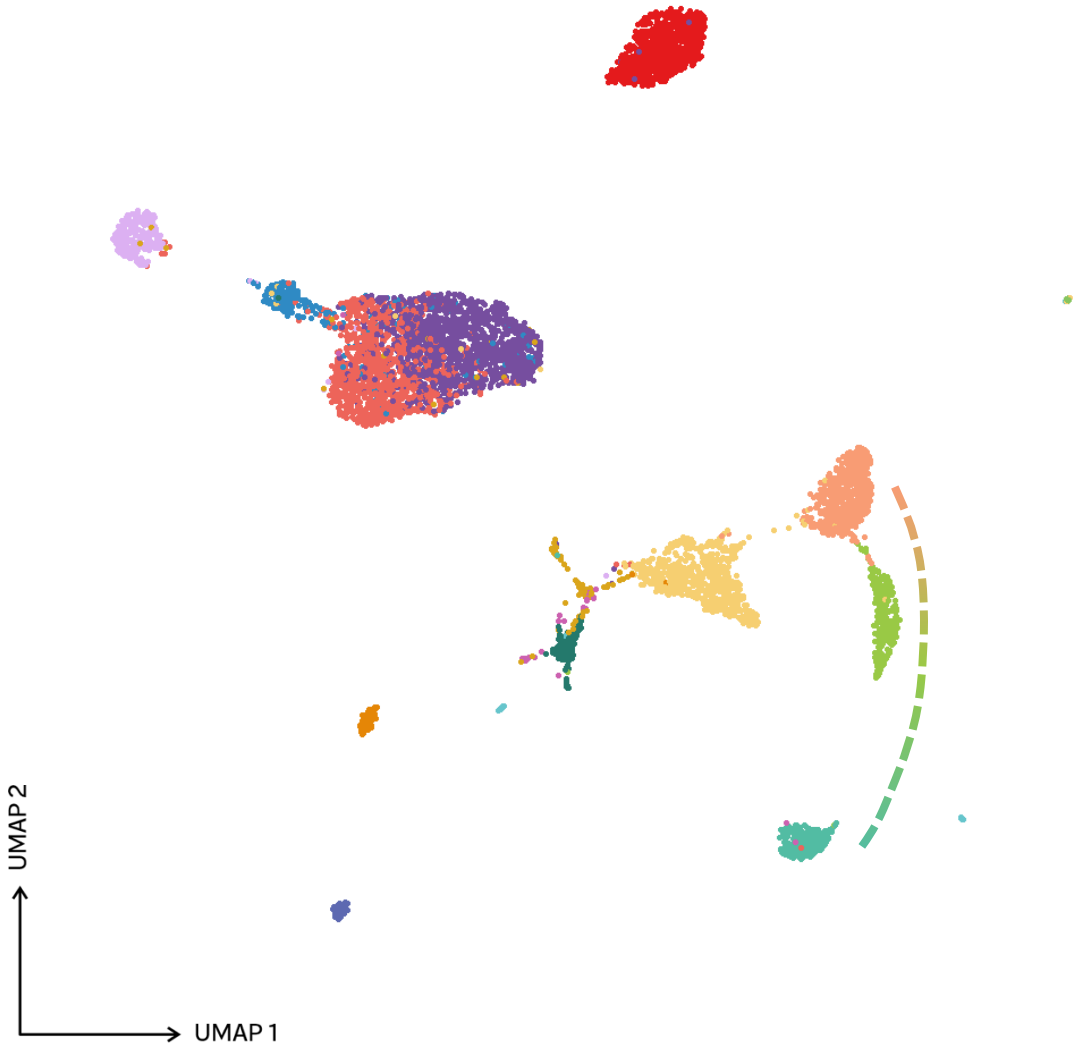


cell types

- Hypothesis: one cell type, one hormone

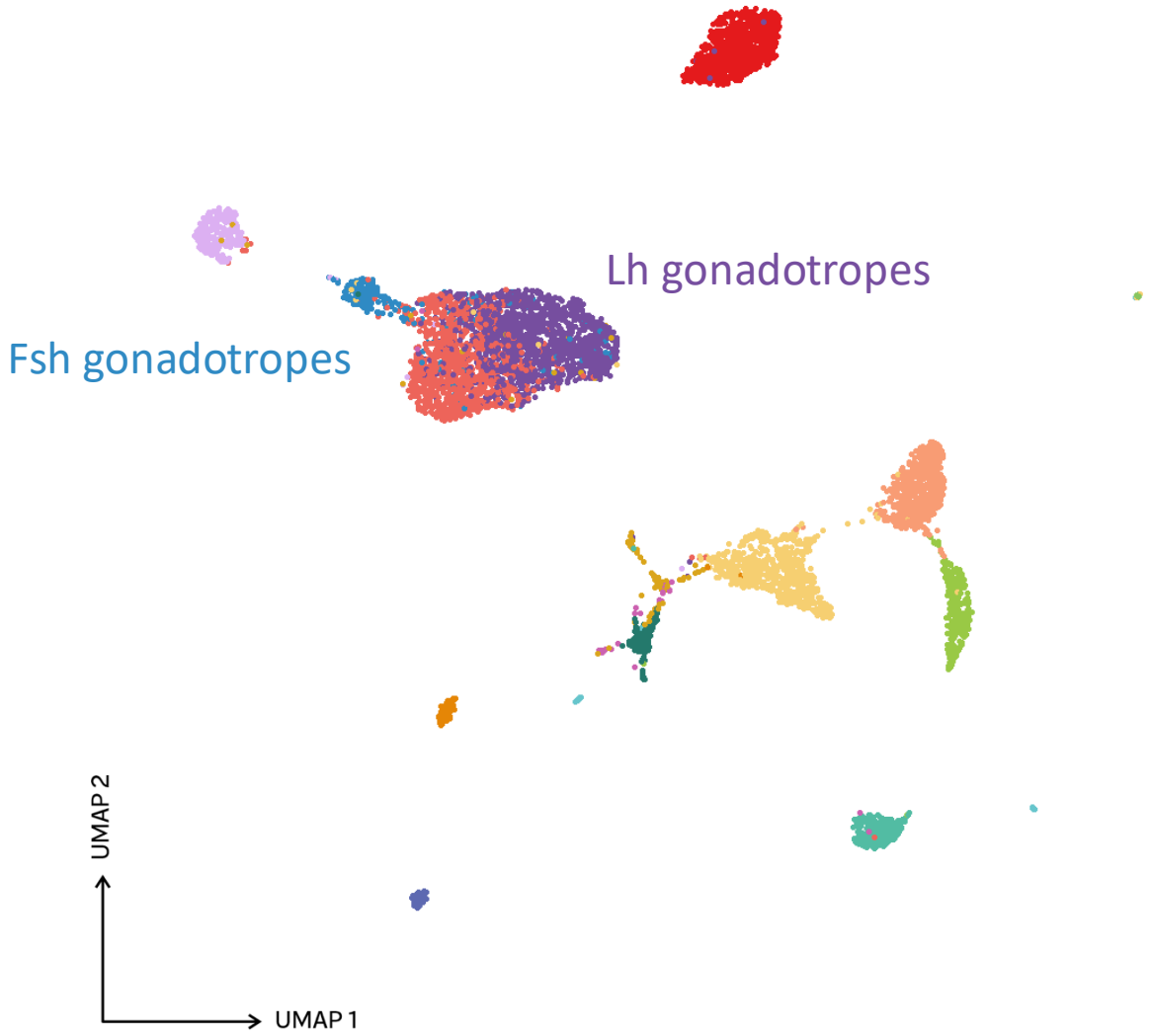
- Adult females (24) and males (23)
- 10x Genomics platform
- 2592/3804 cells
- 57.4K/33.0K reads per cell (median)

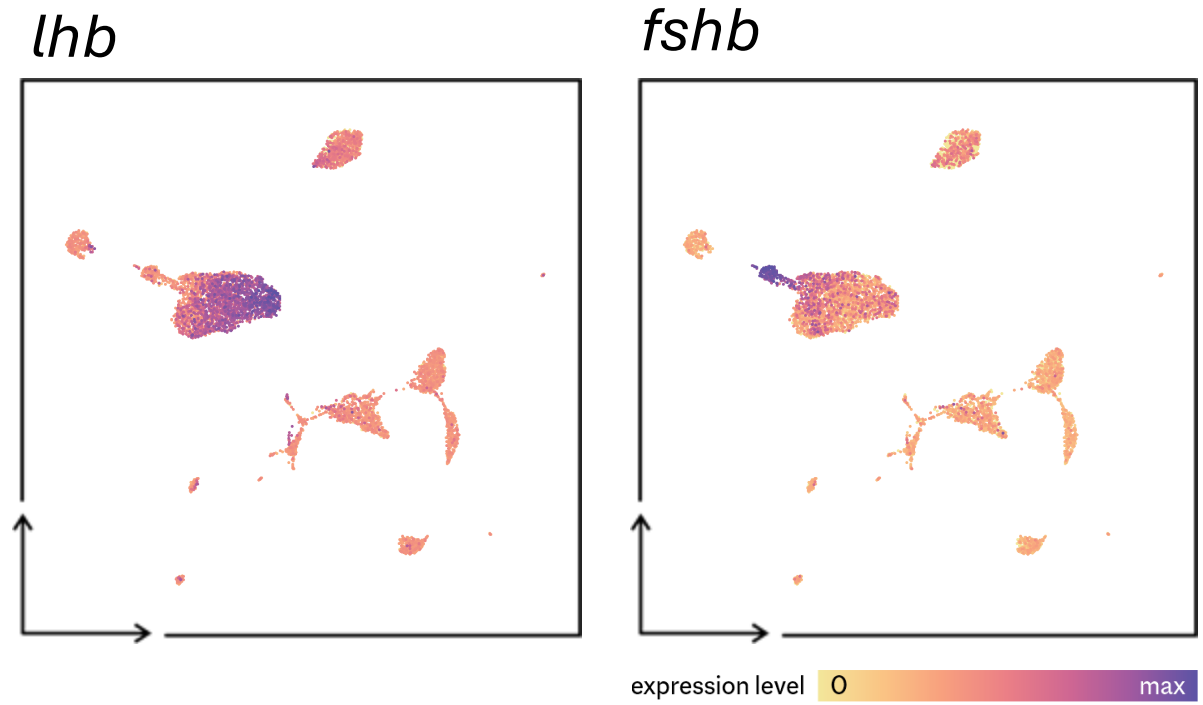
Single-cell RNA-seq: cell clustering



- Every dot is a transcriptome profile
- Visualization in 2 dimensions: similar profiles are close together
- Reveals 16 different cell type clusters
- Suggests differentiation relationships between cell types

Single-cell RNA-seq: cell type assignment



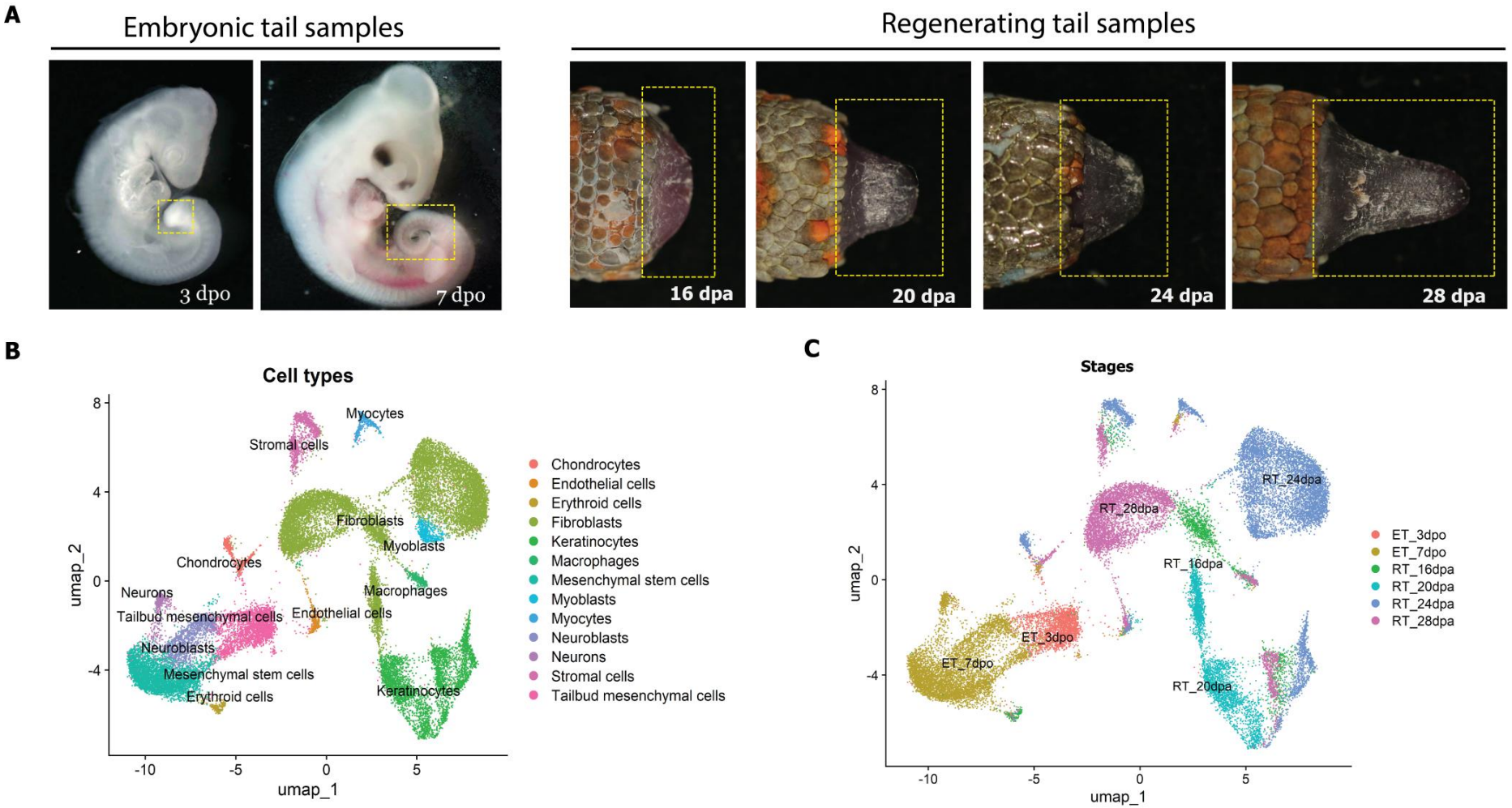


Siddique *et al.* 2021

Single-cell RNA-seq:

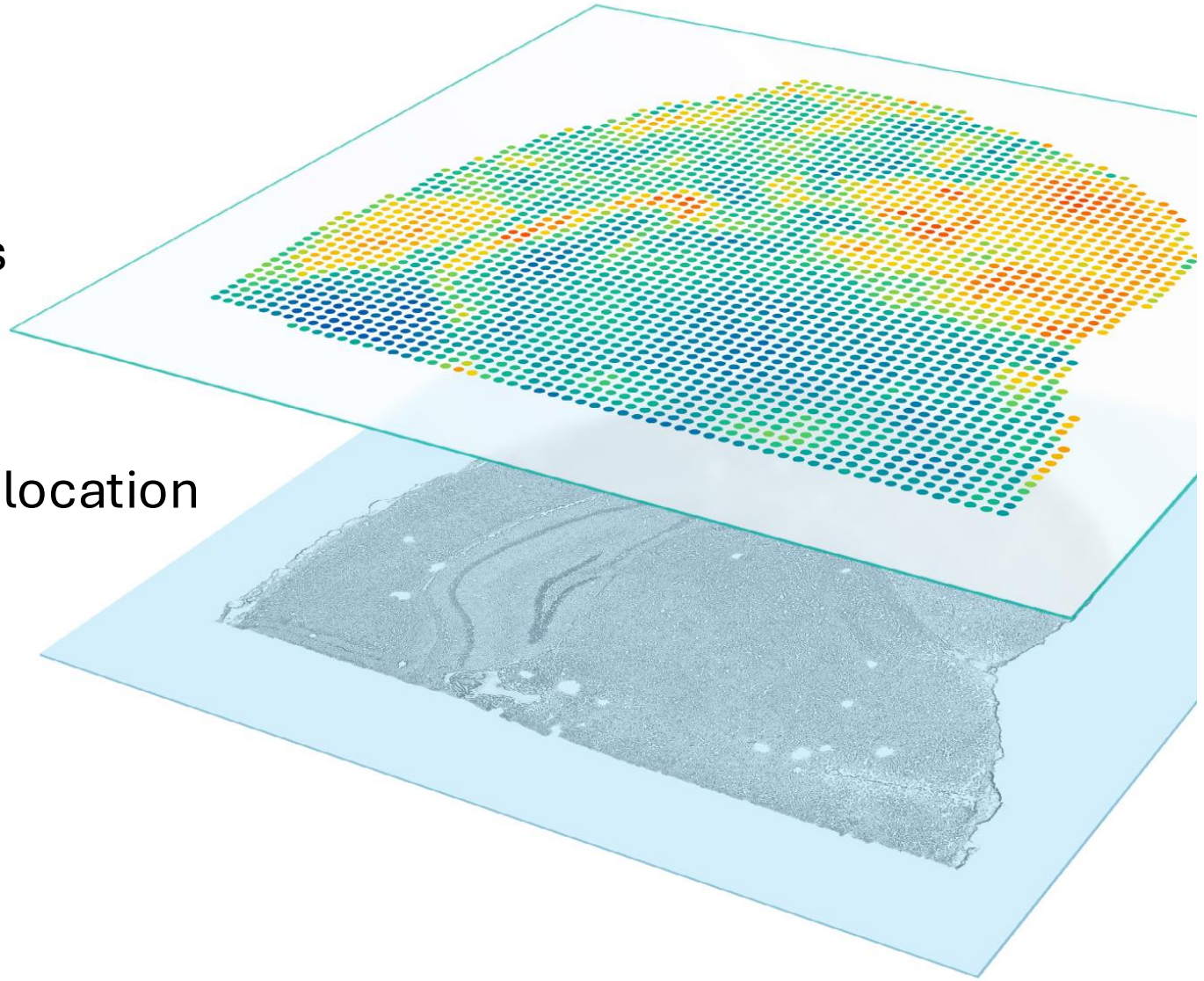


Luthfi Nurhidayat



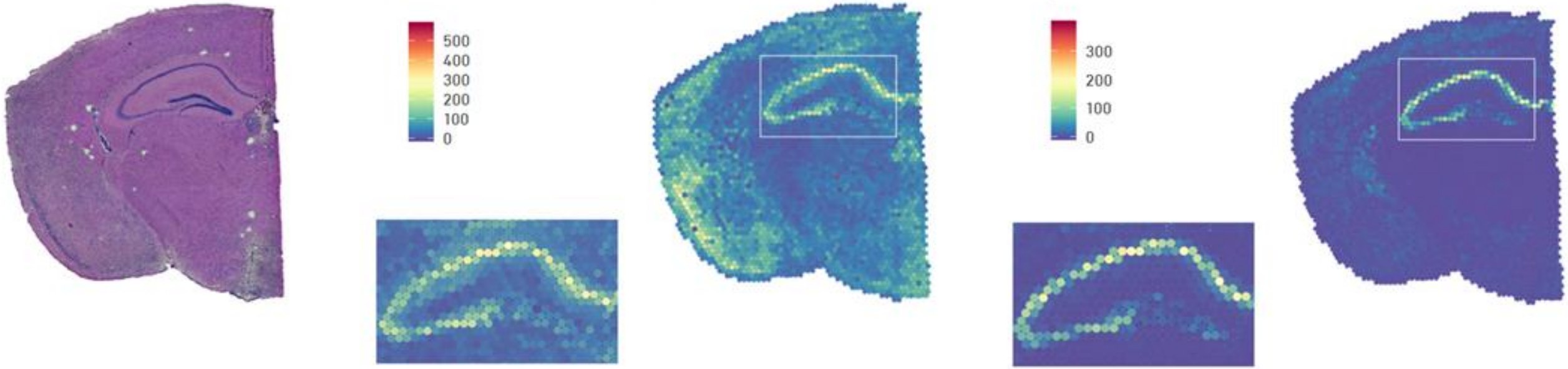
Spatial transcriptomics

- 10x Genomics barcoding on a microscope slide instead of on beads
- 5000 spots, 55 µm diameter
- Barcode allows assignment of a spot location to every transcriptome profile



10x GENOMICS

Spatial transcriptomics



10x GENOMICS

Summary

- RNA-seq provides high-quality gene expression measurements for all genes in a sample
- Single-cell RNA-seq (scRNA-seq) allows classification of cell types
- Spatial transcriptomics combine (single-cell) RNA-seq with *in situ* expression measurements
- Single-cell studies are revolutionizing developmental biology